

Free Riding Under Pressure: A Dynamic Stock–Pressure Model for Public-Good Governance

Summary

Public goods such as environmental governance, open-source maintenance, public infrastructure, and community services share a common dilemma: the collective stock benefits all participants, while contribution costs are borne privately. To explain this mechanism, we develop a simulation-based **Dynamic Stock–Pressure Free-Riding (DSPF) framework**. The framework links individual contribution, public-good stock, maintenance pressure, demand feedback, capacity saturation, and policy intervention in one dynamic multi-agent system. All numerical inputs and outputs are explicitly labeled as **synthetic or simulation-based**, and the random seed is fixed at **20260614** to ensure reproducibility.

For Model I, we construct a stock–pressure dynamic system in which heterogeneous agents contribute effective effort to a shared public-good stock, while natural decay and demand-driven pressure reduce system performance. Agents differ in benefit coefficients, contribution costs, contribution efficiencies, pressure sensitivities, and initial behavioral types. Four controlled scenarios are designed to represent different governance environments: **small volunteer provision**, **rapid growth**, **critical infrastructure**, and **burnout-prone provision**.

For Models II and III, we compare decentralized individual rationality with a social benchmark under identical state conditions. The decentralized benchmark is a **Nash-style marginal-response fixed point**, computed by damped updates; the social benchmark is a **stage-wise planner problem**, solved by numerical optimization rather than by an infinite-horizon dynamic program. Under the updated capacity-saturating dynamics, the simulations reveal a systematic **free-riding gap of 0.805–0.860**, an **under-provision ratio of 0.226–0.299**, and a **welfare-loss ratio of 0.101–0.198**. Among the four scenarios, the burnout-prone case produces the most severe welfare loss because high contribution costs and accumulated maintenance pressure jointly amplify the loss from insufficient provision.

For Model IV, we evaluate subsidy, penalty, reputation, matching fund, threshold governance, and combined portfolio policies. In the critical-infrastructure scenario, the baseline long-run welfare is **66.52**. Subsidy, penalty, reputation, threshold governance, and combined portfolio policies all improve welfare and reduce free riding. The **combined portfolio** gives the highest long-run welfare and the strongest pressure reduction, while **reputation** achieves nearly the same welfare at much lower policy cost. Pareto analysis, subsidy–penalty response surfaces, perturbed-scenario tests, behavioral-weight robustness, solver-damping checks, and Monte Carlo experiments show that **welfare, policy cost, pressure relief, fairness, and free-riding reduction cannot be optimized by one instrument alone**. The best tested subsidy–penalty grid point lies on the penalty boundary, so it is interpreted as a screening result rather than as a global prescription.

The model suggests three governance principles: **use low-cost incentive correction before heavy intervention**, **switch to portfolio governance when maintenance pressure is high**, and **select policies by scenario-specific trade-offs rather than by a single welfare score**. These conclusions are mechanism-oriented and conditional on the synthetic simulation design; they should not be interpreted as empirical estimates for a particular public-good system.

Keywords: Public Goods; Free Riding; Dynamic Game; Social-Planner Benchmark; Policy Intervention; Simulation-Based Modeling

Contents

1	Introduction	3
1.1	Background	3
1.2	Restatement of the Problem	3
1.3	Our Work	3
2	Assumptions and Notations	5
2.1	Assumptions and Justifications	5
2.2	Notations	5
3	Data Strategy and Simulation-Based Scenario Design	6
3.1	Synthetic-Only Data Strategy	6
3.2	Heterogeneous Agent Generation	7
3.3	Synthetic Scenario Design	7
4	Model I: Dynamic Stock–Pressure System	8
4.1	Aggregate Effective Contribution	8
4.2	Public-Good Stock Dynamics with Capacity Saturation	8
4.3	Maintenance Pressure and Demand Dynamics	9
4.4	Value Function and Welfare Components	10
5	Model II: Individual Rationality and Nash-Style Fixed Point	10
5.1	Individual Utility	10
5.2	Damped Marginal-Response Update	11
5.3	Free-Rider Classification	13
6	Model III: Social-Planner Benchmark and Welfare Loss	14
6.1	Stage-Wise Social-Planner Benchmark	14
6.2	Individual-Rational and Social-Planner Comparison	15
6.3	Free-Riding Gap, Under-Provision, and Welfare Loss	16
7	Model IV: Policy Intervention Mechanisms	17
7.1	Policy Variables	17
7.2	Policy Cost Function	18
7.3	Marginal Effects of Policy Instruments	19
7.4	Multi-Criteria Evaluation	19
8	Model Verification and Numerical Reliability	20
9	Numerical Experiments and Results	21
9.1	Baseline Dynamic Behavior	21
9.2	Policy Comparison in the Critical-Infrastructure Scenario	23
9.3	Multi-Dimensional Policy Ranking	25
9.4	Pareto Trade-Off Analysis	25
10	Sensitivity, Robustness, and Uncertainty Analysis	27
10.1	One-Dimensional Sensitivity	27
10.2	Subsidy-Penalty Response Surface	28
10.3	Robustness under Perturbed Scenarios	29
10.4	Behavioral-Response and Solver-Damping Robustness	30
10.5	Monte Carlo Uncertainty Analysis	32
11	Policy Recommendations	34
11.1	Recommendation 1: Use Low-Cost Incentive Correction Before Heavy Intervention	34
11.2	Recommendation 2: Switch to Portfolio Governance under High Maintenance Pressure	35
11.3	Recommendation 3: Avoid One-Size-Fits-All Policy	35
12	Team Member Contributions	35
13	Discussion: Strengths, Limitations, and Extensions	36
13.1	Strengths	36
13.2	Limitations	36
13.3	Future Work	36
13.4	Reproducibility and Auditability Note	36
14	Conclusion	37

1 Introduction

1.1 Background

Public goods share a simple but persistent conflict: their benefits spread across many agents, while their contribution costs are borne individually. Classical public-good theory therefore predicts inefficient private provision whenever benefits are non-excludable and diffuse.^[1, 2, 3] Environmental governance, open-source software maintenance, public infrastructure, and community services all follow this structure. If an agent can enjoy the common outcome while contributing little, waiting for others becomes individually attractive. The resulting strategic gap between private behavior and social provision is the free-riding problem studied in this paper.

The conflict is also dynamic. Public-good quality is not produced once and then held fixed; it decays, responds to demand, and carries maintenance pressure from one period to the next. A static game can explain weak private incentives, but it cannot track how small contribution gaps accumulate into stock shortage, pressure growth, and later welfare loss. The commons-governance literature further indicates that monitoring, sanctions, and local rules may reshape contribution behavior, not only the physical stock.^[4, 5] We therefore treat public-good provision as a dynamic multi-agent system with heterogeneous agents and repeated effort decisions.

1.2 Restatement of the Problem

The task is to build a dynamic supply model for a public good with multiple participating agents. The model must describe stock evolution, individual contribution, natural decay, benefit, cost, and free-riding behavior; compare individual-rational supply with a social-planner benchmark; and then evaluate possible interventions and management rules. Since the assignment emphasizes mechanisms rather than empirical forecasting, we use transparent synthetic settings, parameter experiments, and sensitivity analysis, with all assumptions stated explicitly.

Accordingly, we answer four questions:

1. How can heterogeneous agents and dynamic public-good stock be represented in one model?
2. Does a Nash-style marginal-response fixed point produce lower contribution than a stage-wise social-planner benchmark under controlled scenario settings?
3. How large are the free-riding gap, under-provision ratio, welfare loss, and maintenance pressure under different synthetic scenarios?
4. Which policy instruments can reduce free riding without creating excessive policy cost, inequality, or long-run instability?

1.3 Our Work

We organize the solution around a simulation-based *Dynamic Stock–Pressure Free-Riding* (DSPF) framework. In this framework, contribution incentives, public-good stock, maintenance pressure, and policy instruments form one coupled system. The synthetic scenarios are not treated as observations; they are controlled numerical testbeds for isolating the free-riding mechanism. Figure 1 summarizes the modeling route.

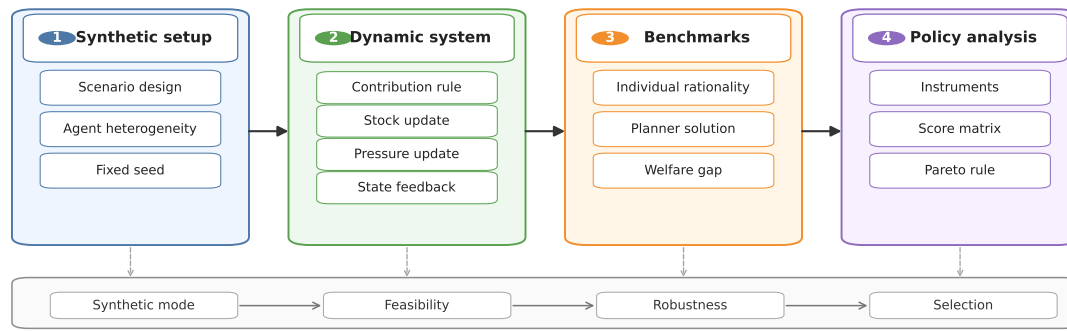


Figure 1: Simulation-based modeling framework. The four main blocks summarize the synthetic testbed, dynamic stock–pressure core, benchmark comparison, and policy-evaluation layer. The lower strip indicates the supporting checks used throughout the study: synthetic-mode interpretation, feasibility, robustness, and final policy selection.

The framework has three layers. The input layer contains the synthetic scenarios and heterogeneous agents. The mechanism layer converts individual effort into effective contribution, then updates public-good stock and maintenance pressure. The decision layer compares decentralized individual rationality with a social-planner benchmark and evaluates policies by long-run indicators. This structure keeps the later experiments traceable to specific mechanisms: Model I defines the stock–pressure dynamics, Model II gives the individual-rational response, Model III measures welfare loss against the planner benchmark, and Model IV compares policy instruments that may correct the marginal-incentive gap.

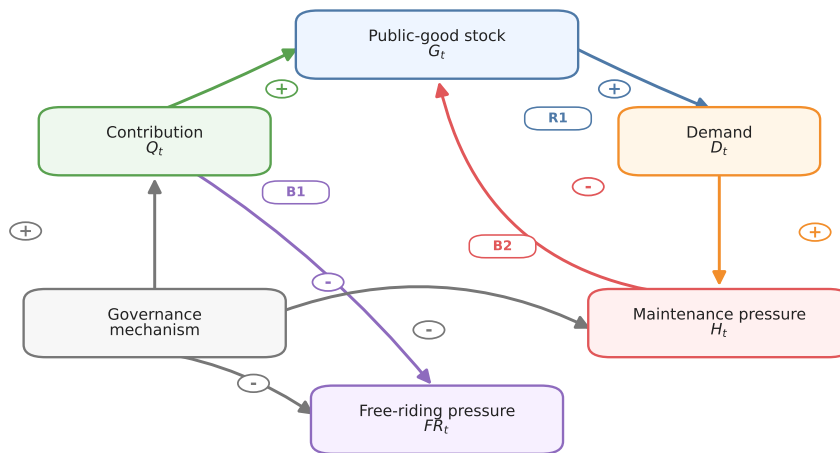


Figure 2: Causal feedback structure of the DSPF framework. Edge signs indicate influence directions represented either in the dynamic equations or in the free-rider diagnostic. Tag R1 marks the stock–demand–pressure feedback path, B1 marks the contribution-to-free-riding correction channel, and B2 marks the maintenance-pressure-to-stock balancing path.

Figure 2 gives a complementary feedback view. Contribution improves stock and relieves pressure; higher stock may attract demand; demand can rebuild pressure; and pressure, if left unchecked, weakens sustainable provision. This loop explains why the experiments report pressure, fairness, policy cost, and stability in addition to contribution and welfare.

Our main work is summarized as follows:

- We construct a DSPF stock-pressure-demand model for public-good provision with heterogeneous agents.
- We compute a Nash-style marginal-response fixed-point proxy and a stage-wise social-planner benchmark under identical synthetic scenarios.
- We quantify free-riding gap, under-provision ratio, welfare loss, maintenance pressure, fairness, and stability.
- We compare six intervention policies and a baseline under multi-criteria evaluation.
- We conduct sensitivity, perturbed-scenario robustness, joint and one-at-a-time behavioral-response robustness, solver-damping, Monte Carlo, and Pareto trade-off analyses to avoid over-interpreting one parameter setting.

2 Assumptions and Notations

2.1 Assumptions and Justifications

Assumption 1: Synthetic scenarios are mechanism-oriented testbeds. The four scenarios are stylized numerical environments rather than real observations. This matches the assignment emphasis on mechanisms, parameter experiments, and sensitivity analysis instead of empirical prediction.

Assumption 2: Agents are heterogeneous but boundedly simple. Each agent is described by a benefit coefficient, contribution cost, contribution efficiency, pressure sensitivity, and initial behavioral type. These attributes retain the key differences among participants while keeping the model interpretable.

Assumption 3: Public-good quality is represented by a normalized stock. The public-good stock is not measured in physical units. It represents the relative quality or service level of a public good, such as infrastructure reliability, software maintainability, or environmental quality.

Assumption 4: The social-planner benchmark is stage-wise. The social planner maximizes current-period welfare given the current state and the next-state transition. The benchmark is therefore a controlled upper reference for the same state, not an infinite-horizon dynamic-programming optimum.

Assumption 5: Nash-style individual rationality is a computational fixed-point proxy. We use damped marginal-response iterations to obtain a numerical fixed point for individual-rational decisions. The behavioral-response weights are calibration values, not empirical estimates; their influence is examined in Section 10.4. Numerical convergence is therefore interpreted as convergence of the implemented response map, not as proof of a unique analytical Nash equilibrium.

Assumption 6: Policy cost is explicitly counted. Interventions are not free. Subsidy, matching fund, reputation operation, public budget injection, backlog reduction, and penalty administration all enter the policy cost function.

2.2 Notations

We separate state and agent variables from policy and evaluation variables. This follows the numerical pipeline: agent and state variables determine the decentralized effort response; policy variables then modify marginal incentives or state transitions; evaluation variables summarize the simulated outcomes. The separation prevents performance indicators from being mistaken for primitive assumptions.

Table 1: State and agent notations used in the dynamic public-good model.

Symbol	Meaning
N, t	Number of agents and discrete time index
$e_{i,t}, e_{\max}$	Effort of agent i and the maximum feasible effort
a_i, b_i, c_i, μ_i	Contribution efficiency, benefit coefficient, cost coefficient, and pressure sensitivity of agent i
$Q_t, Q_{\max}, q(Q_t)$	Aggregate contribution, maximum aggregate contribution, and normalized effective contribution
G_t, H_t, D_t	Public-good stock, maintenance pressure, and demand level at time t
$\bar{G}, \bar{H}, \bar{D}$	Scenario-specific stock, pressure, and demand capacity bounds used in saturation and projection
$\delta, \alpha, \eta, \eta_G, \eta_H$	Natural stock decay, contribution-to-stock conversion, pressure damage, stock-saturation power, and pressure-saturation power
ρ, λ, κ	Natural pressure dissipation, demand-to-pressure conversion, and contribution-based pressure relief
$g_D, \chi, \psi, \varepsilon_t$	Demand growth, stock attraction, pressure deterrence, and simulation noise
$V(G), \omega$	Concave public-good value function and its scaling parameter

Table 2: Policy and evaluation notations.

Symbol	Meaning
s, r, p, m, B, R, τ	Subsidy, reputation, penalty, matching fund, budget injection, backlog reduction, and threshold parameters
$\theta_G, \theta_H, \xi_0, \xi_1, \xi_L$	Behavioral-response weights in the Nash-style marginal-response rule
π, C_{policy}	Policy vector and policy cost
$F_i, FR, \bar{b}_q, \epsilon$	Free-rider indicator, free-riding ratio, high-benefit cutoff, and low-effort threshold
$W_t, \mathbf{e}_t^{SO}$	Social welfare and stage-wise social-planner effort vector
$\Delta FR, \Delta H$	Reduction in free-riding ratio and maintenance pressure relative to the baseline
$\text{Gini}(e)$	Inequality measure of the individual effort distribution

This notation system keeps the model hierarchy explicit. State variables describe the public-good environment, agent variables describe private incentives, policy variables describe intervention levers, and evaluation variables describe comparison criteria. Thus Q_t , G_t , and H_t are generated by effort decisions, while s , p , r , m , B , and R enter only when policy interventions are activated.

3 Data Strategy and Simulation-Based Scenario Design

3.1 Synthetic-Only Data Strategy

We adopt a synthetic-only data route because the course problem calls for mechanism-oriented modeling rather than empirical forecasting. Agent attributes, scenario parameters, and trajectories are generated as controlled numerical testbeds. The purpose is not to estimate a particular community, city, platform, or infrastructure system, but to show how decentralized free riding can generate dynamic under-provision under transparent assumptions.

The numerical design is fixed by seed 20260614, four scenarios, seven policies, a 30-period horizon, 50 Monte Carlo runs, a separate 13×13 pure subsidy–penalty response surface, 80 random portfolios for Pareto analysis, and additional behavioral-weight, scenario-perturbation, and solver-damping checks. In the perturbed-scenario test, the critical-infrastructure population is fixed before decay, cost, and demand changes are imposed; only the stated stress factor changes. Hence the reported welfare, pressure, and free-riding ratios should be read as

controlled simulation outputs, not observed real-world magnitudes.

Table 3 records how the main numerical quantities enter the study. It is an audit table rather than an additional data source: structural assumptions, calibrated behavioral or policy settings, and numerically checked outputs are kept separate.

Table 3: Compact audit of major numerical inputs and calibration choices.

Component	Numerical choice	Audit or interpretation
Reproducibility design	Seed 20260614, $T = 30$, 50 Monte Carlo runs, 80 Pareto samples, and a 13-by-13 response-surface grid	These settings control repeatability and numerical coverage; they are not empirical observations.
Scenario and agent inputs	Four synthetic scenarios; $N = 35$ heterogeneous agents with benefit, cost, efficiency, and pressure-sensitivity variation	Used to separate low pressure, demand growth, infrastructure pressure, and burnout mechanisms.
Behavioral and solver calibration	Nash-style weights (0.90, 0.85, 0.35, 2.80, 0.10); damping 0.42; tolerance 10^{-7} ; L-BFGS-B tolerance 10^{-10}	Treated as synthetic behavioral-response scale settings and numerical choices; convergence, joint-weight, one-at-a-time weight, and damping checks are reported.
Policy and cost settings	Seven policy vectors and synthetic cost weights in C_{policy}	Used for comparative screening of governance mechanisms, not as real policy rates or fiscal estimates.

3.2 Heterogeneous Agent Generation

For each scenario, $N = 35$ agents are generated. Agent benefits are sampled from a log-normal distribution, contribution costs from a gamma distribution, contribution efficiencies from a uniform distribution, and pressure sensitivities from a uniform distribution. Scenario-specific benefit and cost scales adjust the difficulty of provision.

$$\begin{aligned}
 b_i &\sim \text{LogNormal}(\mu_b, \sigma_b) \cdot s_b, \\
 c_i &\sim \text{Gamma}(k_c, \theta_c) \cdot s_c, \\
 a_i &\sim U(a_{\min}, a_{\max}), \\
 \mu_i &\sim U(\mu_{\min}, \mu_{\max}).
 \end{aligned}$$

Initial behavioral types are sampled from three categories: active contributor, conditional contributor, and passive user. These types are not fixed strategies in the equilibrium computation, but they provide an interpretable description of the heterogeneous population.

3.3 Synthetic Scenario Design

Table 4 lists the major parameters of the four synthetic scenarios. The small-volunteer scenario describes a low-pressure voluntary contribution system. The rapid-growth scenario represents increasing demand. The critical-infrastructure scenario has higher public-good importance and maintenance pressure. The burnout-prone scenario combines high cost and high pressure, making under-provision more likely.

Table 4: Synthetic scenario parameters.

Scenario	G_0	H_0	D_0	δ	α	η	λ	κ	Benefit scale	Cost scale
Small volunteer	0.58	0.20	0.30	0.020	0.72	0.16	0.18	0.52	0.80	0.90
Rapid growth	0.60	0.28	0.42	0.038	0.66	0.24	0.40	0.34	1.05	1.03
Critical infrastructure	0.62	0.34	0.58	0.050	0.78	0.30	0.44	0.38	1.35	1.08
Burnout-prone	0.56	0.46	0.48	0.060	0.58	0.36	0.42	0.24	1.00	1.45

These are controlled numerical settings, not empirical observations. Scenario-specific capacities are used in the simulation; the stock- and pressure-saturation powers are fixed at $\eta_G = 2.0$ and $\eta_H = 1.0$.

The scenario parameters are chosen to separate mechanisms rather than fit a particular public-good system. Small-volunteer provision starts from low pressure and moderate cost. Rapid growth raises demand and demand-to-pressure conversion, making delayed pressure

accumulation visible. Critical infrastructure increases benefit scale and pressure damage, reflecting the high cost of service failure. Burnout-prone provision combines high cost, high initial pressure, and weak pressure relief, representing systems in which repeated contribution is hard to sustain. Figure 3 summarizes the four controlled testbeds.

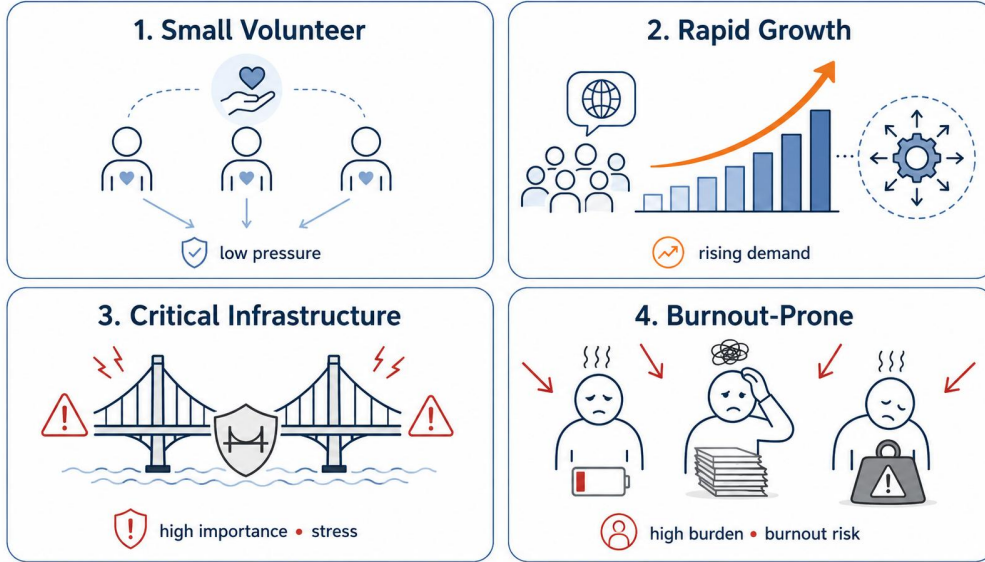


Figure 3: Conceptual scenario taxonomy used in the synthetic experiments. The figure summarizes the dominant mechanism stressed by each testbed: low-pressure volunteer provision, demand-driven growth, infrastructure-like pressure, and burnout-prone contribution burden. It is a conceptual guide to the scenario design, not an empirical data visualization.

4 Model I: Dynamic Stock–Pressure System

4.1 Aggregate Effective Contribution

Each agent chooses a contribution effort $e_{i,t} \in [0, e_{\max}]$. The aggregate effective contribution is

$$Q_t = \sum_{i=1}^N a_i e_{i,t}.$$

Because public-good production often has diminishing returns, we use a logarithmic transformation

$$q(Q_t) = \frac{\log(1 + Q_t)}{\log(1 + Q_{\max})},$$

where $Q_{\max} = \sum_i a_i e_{\max}$. Thus $q(Q_t)$ is a normalized contribution effectiveness term.

4.2 Public-Good Stock Dynamics with Capacity Saturation

The public-good stock evolves through a capacity-constrained saturating transition. First define the stock saturation factor

$$\Gamma_G(G_t) = 1 - \left(\frac{G_t}{\bar{G}} \right)^{\eta_G}, \quad 0 \leq G_t \leq \bar{G}.$$

The unprojected next-period stock is

$$\tilde{G}_{t+1} = (1 - \delta)G_t + \alpha(1 + m)q(Q_t)\Gamma_G(G_t) - \eta H_t + B, \quad (1)$$

and the feasible stock is

$$G_{t+1} = \Pi_{[0, \bar{G}]}(\tilde{G}_{t+1}). \quad (2)$$

Here δ is natural decay, α is the conversion rate from contribution to stock, m is matching-fund intensity, η measures damage from maintenance pressure, B is direct public budget injection, and Π denotes projection to the feasible interval. The saturation factor is the main economic mechanism: when the public-good stock approaches its scenario-specific capacity $\bar{G}$, the marginal stock improvement from additional contribution becomes smaller. The final projection is only a feasibility safeguard.

4.3 Maintenance Pressure and Demand Dynamics

Maintenance pressure is also modeled as a bounded dynamic state. Let

$$\Gamma_H(H_t) = 1 - \left(\frac{H_t}{\bar{H}} \right)^{\eta_H}, \quad 0 \leq H_t \leq \bar{H}.$$

The pressure transition is

$$\tilde{H}_{t+1} = (1 - \rho)H_t + \lambda D_t \Gamma_H(H_t) - \kappa q(Q_t) - R, \quad (3)$$

$$H_{t+1} = \Pi_{[0, \bar{H}]}(\tilde{H}_{t+1}), \quad (4)$$

where ρ is natural pressure dissipation, λ is demand-to-pressure conversion, κ is contribution-based pressure relief, and R is direct backlog reduction. Free riding therefore affects pressure through insufficient aggregate contribution and weaker pressure relief, rather than through a separate ad hoc free-riding term.

Demand evolves as

$$D_{t+1} = D_t \left(1 + g_D + \chi \frac{G_t}{\bar{G}} - \psi \frac{H_t}{\bar{H}} \right) + \varepsilon_t, \quad (5)$$

with final projection to the scenario-specific demand range. The term g_D is baseline demand growth. Better public-good stock may attract more demand, while higher maintenance pressure may reduce demand or effective usage. The noise term ε_t is used only in simulation, and its scale is specified by the synthetic scenario.

Proposition 1 (Feasible state invariance). Suppose the initial state satisfies $G_0 \in [0, \bar{G}]$, $H_0 \in [0, \bar{H}]$, and $D_0 \in [0, \bar{D}]$. If the stock, pressure, and demand updates are followed by their stated interval projections, then every simulated trajectory satisfies

$$G_t \in [0, \bar{G}], \quad H_t \in [0, \bar{H}], \quad D_t \in [0, \bar{D}]$$

for all periods t , for any feasible effort vector and any policy vector used in the experiments.

Proof. For any closed interval $[L, U]$, the projection operator satisfies $\Pi_{[L, U]}(x) \in [L, U]$ for every real number x . Equations 2 and 4 apply this operator to the raw stock and pressure updates; the demand update is treated in the same projected manner in the simulation. Therefore the next state remains inside the feasible interval whenever the current state is simulated. The claim follows by induction over t . $\square$

The dynamic equations give two useful local monotonicity relations. Since

$$\frac{\partial q(Q_t)}{\partial e_{i,t}} = \frac{a_i}{(1 + Q_t) \log(1 + Q_{\max})} > 0,$$

we have, at interior points away from projection boundaries,

$$\frac{\partial \tilde{G}_{t+1}}{\partial e_{i,t}} = \alpha(1 + m)\Gamma_G(G_t) \frac{a_i}{(1 + Q_t) \log(1 + Q_{\max})} \geq 0,$$

$$\frac{\partial \tilde{H}_{t+1}}{\partial e_{i,t}} = -\kappa \frac{a_i}{(1 + Q_t) \log(1 + Q_{\max})} < 0.$$

Individual effort therefore has two system-level effects: it raises next-period stock when capacity is not binding and lowers next-period maintenance pressure. These signs form the technical basis for the marginal externality derived in Model II.

4.4 Value Function and Welfare Components

Public-good stock creates utility through a concave value function

$$V(G) = \log(1 + \omega G),$$

where $\omega = 3.6$ in the numerical experiments. Concavity reflects diminishing marginal benefit from stock improvement.

Equations 1–5 define the dynamic core of the DSPF framework. They also show why a one-period public-good game is insufficient: current effort changes both next-period stock and maintenance burden, which then affect future welfare and policy needs. Section 9.1 reports the numerical behavior generated by this core.

5 Model II: Individual Rationality and Nash-Style Fixed Point

5.1 Individual Utility

An individual agent trades off public-good benefit, contribution cost, pressure exposure, and policy incentives. The setup follows the standard game-theoretic idea that each agent chooses a feasible action while treating the other agents' actions and the current state as given.^[6, 7] Given next-period stock and pressure, the utility of agent i is

$$\begin{aligned} u_i(e_i, e_{-i}; \pi) = & b_i V(G_{t+1}) - \frac{1}{2} c_i e_i^2 - \mu_i H_{t+1} + s e_i + r \log(1 + e_i) \\ & + p \mathbf{1}_{\{b_i \geq \bar{b}_q\}} \left(0.35 e_i - 1.4 (\tau^* - e_i)_+^2 \right) - 0.10 p (\tau^* - e_i)_+. \end{aligned}$$

Here s is subsidy intensity, r is reputation intensity, p is penalty intensity, $\bar{b}_q$ is the high-benefit cutoff, and $\tau^* = \max(\tau, \epsilon)$ is the implemented threshold. The two penalty-related terms are written separately to make their scope explicit: high-benefit agents receive the targeted threshold incentive, while every below-threshold agent receives a smaller generic threshold-push term. The structure directly represents the free-riding incentive: high-benefit agents may obtain public-good benefit even when their own effort is low.

The marginal incentive gap becomes clear by comparing the individual and social first-order conditions. Ignoring boundary projection and non-smooth threshold terms for exposition, agent i responds to

$$\frac{\partial u_i}{\partial e_i} \approx b_i V'(G_{t+1}) \frac{\partial G_{t+1}}{\partial e_i} - c_i e_i - \mu_i \frac{\partial H_{t+1}}{\partial e_i} + \text{policy marginal terms}, \quad (6)$$

whereas the social planner internalizes the effect of agent i 's effort on all agents:

$$\frac{\partial W}{\partial e_i} \approx \sum_{j=1}^N b_j V'(G_{t+1}) \frac{\partial G_{t+1}}{\partial e_i} - c_i e_i - \sum_{j=1}^N \mu_j \frac{\partial H_{t+1}}{\partial e_i} - \frac{\partial C_{\text{policy}}}{\partial e_i}. \quad (7)$$

The private condition contains only b_i and μ_i , while the social condition contains aggregate benefit and aggregate pressure exposure. The omitted spillover is the mathematical source of under-contribution in the decentralized benchmark.

Proposition 2 (Local marginal source of free riding). At an interior baseline point away from projection boundaries and threshold discontinuities, the decentralized marginal incentive omits the positive spillover from agent i 's effort to the other agents. Therefore, when the remaining agents have positive aggregate benefit or pressure exposure, the social marginal value of effort is larger than the private marginal value before policy correction.

Proof. For an interior baseline point, Equations 1 and 3 imply

$$\begin{aligned}\frac{\partial \tilde{G}_{t+1}}{\partial e_i} &= \alpha(1+m)\Gamma_G(G_t) \frac{a_i}{(1+Q_t)\log(1+Q_{\max})} \geq 0, \\ \frac{\partial \tilde{H}_{t+1}}{\partial e_i} &= -\kappa \frac{a_i}{(1+Q_t)\log(1+Q_{\max})} < 0.\end{aligned}$$

Subtracting the private marginal condition in Equation 6 from the social marginal condition in Equation 7 gives

$$\begin{aligned}\frac{\partial W}{\partial e_i} - \frac{\partial u_i}{\partial e_i} &= \left(\sum_{j \neq i} b_j \right) V'(G_{t+1}) \frac{\partial \tilde{G}_{t+1}}{\partial e_i} \\ &\quad - \left(\sum_{j \neq i} \mu_j \right) \frac{\partial \tilde{H}_{t+1}}{\partial e_i} - \frac{\partial C_{\text{policy}}}{\partial e_i} - \text{private policy terms}.\end{aligned}$$

For the baseline comparison, the direct policy terms vanish. The first spillover term is nonnegative because increasing e_i raises next-period public-good stock for other beneficiaries unless the capacity boundary is already binding. The second spillover term is positive because $\partial \tilde{H}_{t+1}/\partial e_i < 0$, so the same effort reduces pressure exposure for other agents. Thus the private response ignores positive marginal effects on others, which explains why the individual-rational response tends to stop before the social-planner benchmark under the synthetic baseline settings. $\square$

5.2 Damped Marginal-Response Update

The decentralized solution is obtained by damped marginal-response updates. At each iteration, an agent responds to marginal public benefit, marginal pressure relief, private cost, and policy incentives. To make the numerical calibration explicit, we write the raw marginal response with five behavioral-response weights:

$$\begin{aligned}\widetilde{MR}_i^{(k)} &= \frac{1}{c_i} \left[\theta_G a_i b_i \alpha(1+m)\Gamma_G(G_t) V'(G_t) q'(Q^{(k)}) + \theta_H a_i \mu_i \kappa q'(Q^{(k)}) + s + \frac{r}{1+e_i^{(k)}} \right. \\ &\quad \left. + p \left(\xi_0 + \xi_1 (\tau^* - e_i^{(k)})_+ \right) \mathbf{1}_{\{b_i \geq \bar{b}_q\}} + \xi_L p \mathbf{1}_{\{e_i^{(k)} < \tau^*\}} \right].\end{aligned}$$

Here $q'(Q) = \{(1+Q)\log(1+Q_{\max})\}^{-1}$, $\Gamma_G(G_t)$ is the stock saturation factor, and $\tau^* = \max(\tau, \epsilon)$ is the implemented penalty threshold. The baseline values are

$$(\theta_G, \theta_H, \xi_0, \xi_1, \xi_L) = (0.90, 0.85, 0.35, 2.80, 0.10).$$

These constants are controlled behavioral-response scale settings rather than empirical estimates. The value 0.90 weights the perceived stock-benefit channel, 0.85 weights maintenance-pressure relief, 0.35 represents the base targeted deterrence for high-benefit low-effort agents, 2.80 controls the threshold-gap penalty response, and 0.10 gives a mild push to agents below the contribution threshold. Their role is to place stock benefit, pressure relief, targeted penalty, and threshold-push channels on comparable effort-response scales inside a synthetic model; none

of them is fitted to observed behavior. Table 5 records this interpretation and the corresponding audit boundary. Because these weights are calibrated modeling choices, Section 10.4 perturbs them jointly, perturbs them one at a time, and also varies the solver damping rate.

The audit also clarifies the policy experiments. The first two weights operate whenever stock benefit and pressure relief enter the private response; the three penalty-related weights operate only when a penalty or threshold instrument is active. Thus the baseline comparison tests decentralized under-provision without punishment, while the intervention experiments add targeted deterrence to the same response rule.

Table 5: Audit and robustness treatment of the Nash-style behavioral-response weights.

Baseline	Symbol	Channel	Calibration role and interpretation boundary	OAT tested range
0.90	θ_G	Perceived stock-benefit response	Keeps the private stock-benefit channel on the same effort-response scale as cost and pressure channels; not an empirical estimate.	0.630–1.170
0.85	θ_H	Perceived pressure-relief response	Keeps the maintenance-pressure channel comparable to the stock-benefit channel without estimating real behavioral perception.	0.595–1.105
0.35	ξ_0	Base targeted deterrence	Sets the baseline upward push generated by targeted penalty incentives for high-benefit low-effort agents.	0.245–0.455
2.80	ξ_1	Threshold-gap penalty response	Controls how strongly the marginal response increases when effort is far below the threshold.	1.960–3.640
0.10	ξ_L	Mild threshold push	Gives a weak generic push to below-threshold agents; intentionally smaller than the targeted penalty-gap channel.	0.070–0.130

OAT means one-at-a-time perturbation. All weights are synthetic calibration values. The penalty-related weights are active only when the penalty intensity is nonzero.

The projected and damped update is

$$e_i^{(k+1)} = (1 - \omega_B)e_i^{(k)} + \omega_B \Pi_{[0, e_{\max}]} \left(\widetilde{MR}_i^{(k)} \right), \quad \omega_B = 0.42,$$

where Π denotes projection onto the feasible interval. This is why the paper refers to the output as a Nash-style individual-rational response benchmark instead of a proven unique Nash equilibrium. The computation is used to construct a reproducible decentralized decision baseline; it does not assert analytical uniqueness or global equilibrium selection. Section 10.4 also varies ω_B to check that the numerical proxy is not driven by a single damping choice.

Table 6: Computational procedure for the Nash-style individual-rational solution.

Step	Operation	Description
1	Initialize	Set a feasible initial effort vector $\mathbf{e}^{(0)} \in [0, e_{\max}]^N$.
2	Individual update	For each agent i , compute a marginal response MR_i from the private first-order balance and project it to $[0, e_{\max}]$.
3	Damping	Update $\mathbf{e}^{(k+1)} = (1 - \omega_B)\mathbf{e}^{(k)} + \omega_B \Pi(BR(\mathbf{e}^{(k)}))$ with baseline $\omega_B = 0.42$.
4	Stopping rule	Stop when $\max_i e_i^{(k+1)} - e_i^{(k)} $ is below the tolerance or when the maximum iteration number is reached.

In the numerical calculation, the initial effort is 0.06 for every agent, the damping weight on the raw projected response is 0.42, the tolerance is 10^{-7} , and the maximum number of damped

sweeps is 60. Under this setting, all four baseline scenarios converge before the sweep limit. We still call the output a Nash-style individual-rational response benchmark rather than a unique Nash equilibrium, because the computation verifies convergence of the implemented marginal-response map but does not prove analytical uniqueness or rule out other theoretical equilibria. Table 7 reports the convergence diagnostics. The later damping check further confirms that the aggregate benchmark is nearly unchanged when the raw-response weight is varied around the baseline value.

Table 7: Convergence check for the baseline Nash-style marginal-response update.

Scenario	Converged	Damped sweeps	Max residual	Q^{IR}	Mean effort
Small volunteer	Yes	27	5.6×10^{-17}	2.688	0.073
Rapid growth	Yes	28	1.7×10^{-16}	3.344	0.100
Critical infrastructure	Yes	27	$< 10^{-16}$	3.007	0.089
Burnout-prone	Yes	27	7.5×10^{-15}	1.601	0.046

The maximum sweep limit is 60 and the stopping tolerance is 10^{-7} . The residuals are at machine precision after convergence of the implemented damped marginal-response map.

5.3 Free-Rider Classification

An agent is classified as a free rider if it has high benefit but contributes below a low-effort threshold. Let $\bar{b}_q$ be the high-benefit quantile cutoff and let ϵ be the free-rider effort threshold. Then

$$F_i = \mathbf{1}\{e_i < \epsilon, b_i \geq \bar{b}_q\}.$$

The free-riding ratio is

$$FR = \frac{1}{N} \sum_{i=1}^N F_i.$$

In the reported simulation setting, $\epsilon = 0.08$ and $q = 0.55$, so an agent must be both below the low-effort threshold and above the 55th percentile of the benefit distribution to be classified as a free rider. This definition avoids labeling low-benefit agents as free riders simply because they contribute little. It also makes the indicator conservative: passive agents with little expected benefit are not counted as free riders unless they are among the higher-benefit participants.

For policy evaluation, this definition is more informative than a pure low-effort rule. A pure low-effort threshold would label many capacity-constrained or low-benefit participants as governance failures even when their private gain from the public good is small. The high-benefit condition focuses the indicator on agents who benefit substantially from the public good but still contribute little, which is the strategic free-riding behavior of interest.

Figure 4 checks this definition on the synthetic population. The distribution is not an estimate of real behavior; it shows that the rule separates low-effort, high-benefit agents from low-effort agents with limited private benefit. This distinction matters because the policy mechanisms target strategic free riding, not low contribution alone.

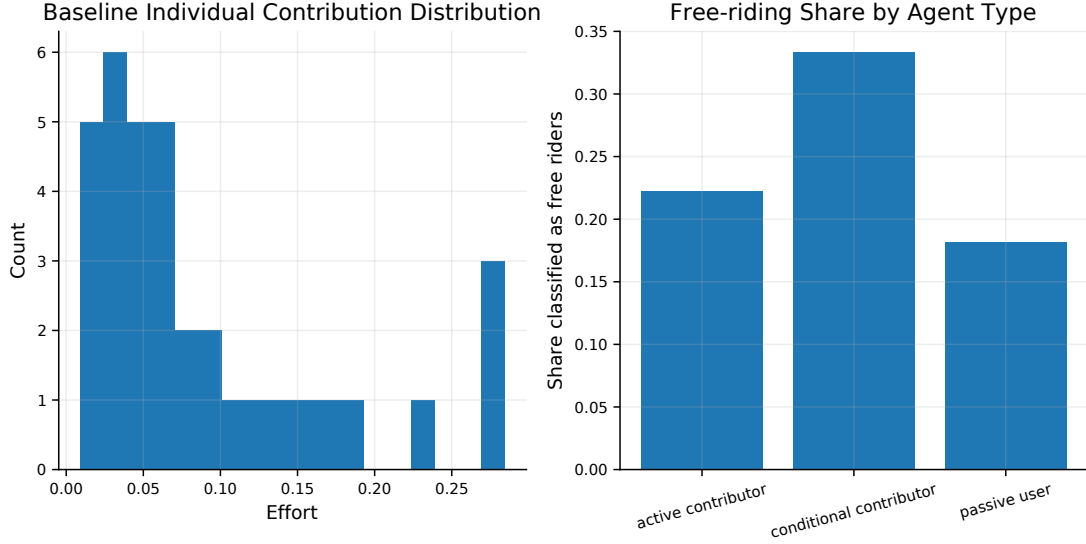


Figure 4: Synthetic contribution and benefit distribution used for free-rider classification. The figure illustrates how the model distinguishes high-benefit low-effort free riders from low-benefit passive agents.

6 Model III: Social-Planner Benchmark and Welfare Loss

6.1 Stage-Wise Social-Planner Benchmark

The social planner chooses the vector of efforts to maximize total welfare after accounting for policy cost:

$$W(\mathbf{e}_t) = \sum_{i=1}^N \left[b_i V(G_{t+1}) - \frac{1}{2} c_i e_{i,t}^2 - \mu_i H_{t+1} \right] - C_{\text{policy}}.$$

The stage-wise social-planner benchmark is

$$\mathbf{e}_t^{SO} = \arg \max_{\mathbf{e}_t \in [0, e_{\max}]^N} W(\mathbf{e}_t).$$

Proposition 3 (Existence of the stage-wise planner benchmark). For every fixed synthetic scenario state (G_t, H_t, D_t) , the baseline stage-wise planner problem used in the individual-rational versus social-planner comparison has at least one feasible maximizer $\mathbf{e}_t^{SO} \in [0, e_{\max}]^N$.

Proof. The feasible effort set $[0, e_{\max}]^N$ is closed and bounded, hence compact. For a fixed state and fixed agent attributes, the aggregate contribution $Q_t = \sum_i a_i e_{i,t}$ and the normalized effectiveness $q(Q_t)$ are continuous in $\mathbf{e}_t$. The unprojected stock and pressure transitions are therefore continuous functions of $\mathbf{e}_t$, and projection onto closed intervals preserves continuity. Since $V(G) = \log(1 + \omega G)$, the benefit term, quadratic effort cost, and pressure term in the baseline welfare function are continuous in $\mathbf{e}_t$. By the extreme value theorem, the continuous welfare function attains a maximum on the compact feasible set. Therefore a stage-wise planner maximizer exists. This proposition only establishes existence of a finite-dimensional benchmark; it does not imply uniqueness or solve an infinite-horizon social optimum. $\square$

We solve the planner problem with L-BFGS-B under box constraints, using maximum iteration number 120 and function tolerance 10^{-10} . The result is a tractable upper reference for the current state, not an infinite-horizon dynamic-programming solution. Since the transition includes saturation and interval projection, the benchmark is interpreted as the output of a specified numerical procedure. This is sufficient for the present purpose: diagnosing the direction and magnitude of decentralized under-provision under comparable state conditions.

In all four scenarios, the planner benchmark gives higher contribution and higher welfare than the Nash-style solution. Proposition 3 ensures that the finite-dimensional baseline benchmark is well-defined; uniqueness and infinite-horizon optimality are not claimed.

6.2 Individual-Rational and Social-Planner Comparison

Table 8 and Figure 5 compare the Nash-style individual-rational benchmark, denoted by IR , with the stage-wise social-planner benchmark under the capacity-saturating dynamics. In every synthetic scenario, aggregate effective contribution is much lower under individual rationality. This reproduces the classical public-good mechanism: agents bear their own contribution cost but internalize only part of the shared benefit.^[2, 3]

Table 8: Nash-style individual rationality versus stage-wise social-planner benchmark under synthetic scenarios.

Scenario	Q^{IR}	Q^{SO}	G^{IR}	G^{SO}	Free-riding gap	Under-provision	Welfare loss
Small volunteer	2.688	14.069	0.743	0.966	0.809	0.231	0.101
Rapid growth	3.344	17.149	0.728	0.940	0.805	0.226	0.116
Critical infrastructure	3.007	17.453	0.737	1.013	0.828	0.272	0.147
Burnout-prone	1.601	11.455	0.488	0.695	0.860	0.299	0.198

The benchmark is stage-wise and simulation-based. It is used to diagnose under-provision under identical state conditions, not to claim an infinite-horizon global optimum.

Table 8 reports one-step benchmark outcomes under an identical current state. It is therefore used for mechanism diagnosis rather than for long-run policy ranking. Later policy tables report tail averages from dynamic trajectories, so their welfare numbers should not be compared mechanically with the one-step values in this benchmark table. The qualitative conclusion is unchanged after introducing capacity saturation: decentralized contribution remains far below the social-planner benchmark in all four scenarios.

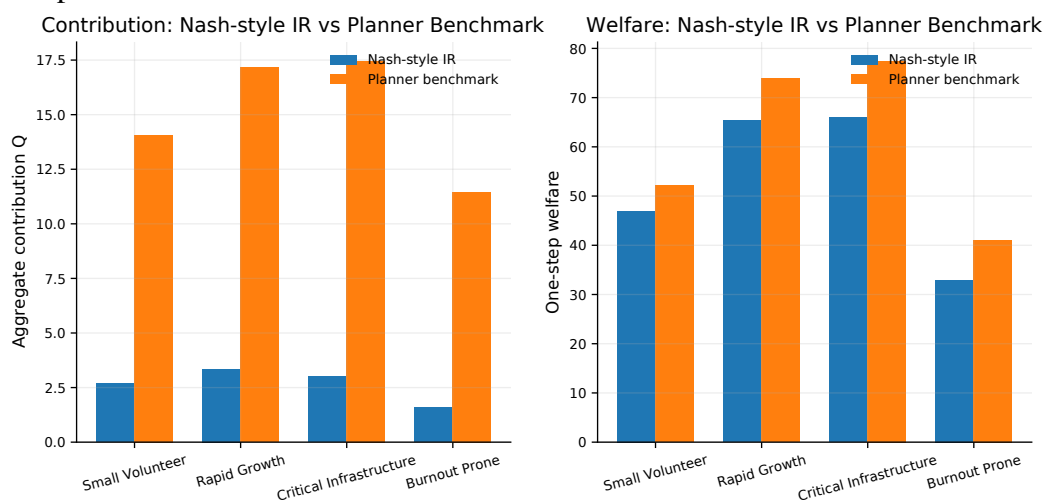


Figure 5: Nash-style individual rationality versus social-planner benchmark. Controlled numerical experiments show lower contribution and welfare under decentralized individual rationality.

The table gives the numerical gap, and the figure makes its direction immediate. Across all scenarios, the planner selects a much higher aggregate effective contribution than the individual-rational response. The reason is the same in each case: the private agent bears full marginal cost but receives only a fraction of the social marginal benefit.

The welfare gap is not determined by contribution alone. The burnout-prone scenario has both the highest free-riding gap and the highest welfare loss, yet the loss also depends on cost and

pressure dynamics. A policy that raises contribution may still be unattractive if it is expensive or leaves pressure unresolved. For this reason, later policy comparisons include welfare, cost, free-riding reduction, pressure, fairness, and stability.

Dynamic Comparison between Nash-style IR and Planner Benchmark (Critical Infrastructure)

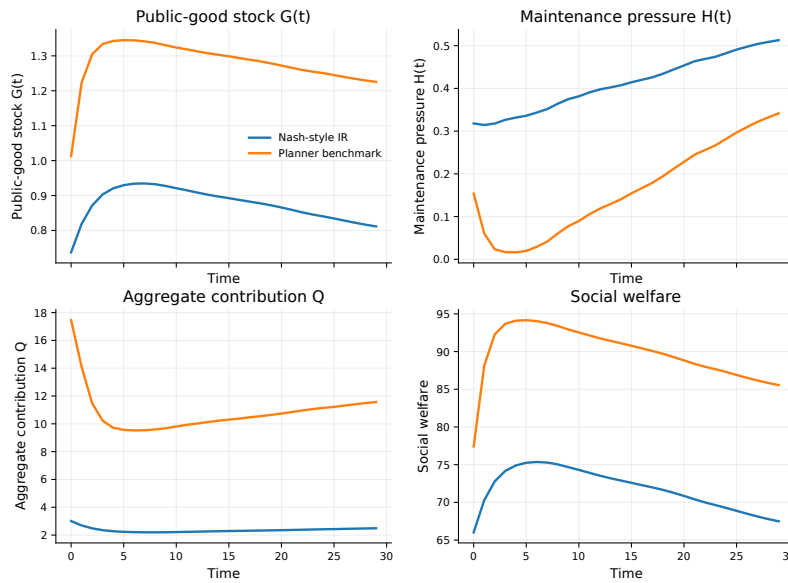


Figure 6: Dynamic trajectory comparison between Nash-style individual rationality and the stage-wise social-planner benchmark. The trajectory view shows how under-provision accumulates through repeated contribution decisions.

Figure 6 extends the one-step comparison to a dynamic trajectory. The planner does not simply raise the initial contribution level; it changes the path of stock accumulation and pressure relief over time. This view is closer to the original problem, since free riding becomes costly through repeated under-maintenance rather than through one isolated decision.

6.3 Free-Riding Gap, Under-Provision, and Welfare Loss

We define the three main diagnostic measures as

$$\begin{aligned} \text{FreeRidingGap} &= \frac{Q^{SO} - Q^{IR}}{Q^{SO}}, \\ \text{UnderProvision} &= \frac{G^{SO} - G^{IR}}{G^{SO}}, \\ \text{WelfareLoss} &= \frac{W^{SO} - W^{IR}}{|W^{SO}|}. \end{aligned}$$

The synthetic scenarios produce a free-riding gap between 0.805 and 0.860, an under-provision ratio between 0.226 and 0.299, and a welfare loss ratio between 0.101 and 0.198. Figure 7 decomposes the gap by scenario and shows that the burnout-prone scenario has the largest welfare loss.

The three ratios measure different layers of the mechanism. The free-riding gap compares contribution levels. The under-provision ratio translates that decision gap into stock shortage. The welfare-loss ratio combines benefit, cost, and pressure effects. A scenario can therefore have a large contribution gap but a smaller welfare loss if the stock system absorbs part of the shortage; conversely, high pressure can magnify a comparable contribution gap. This is why we report the decomposition instead of a single aggregate score.

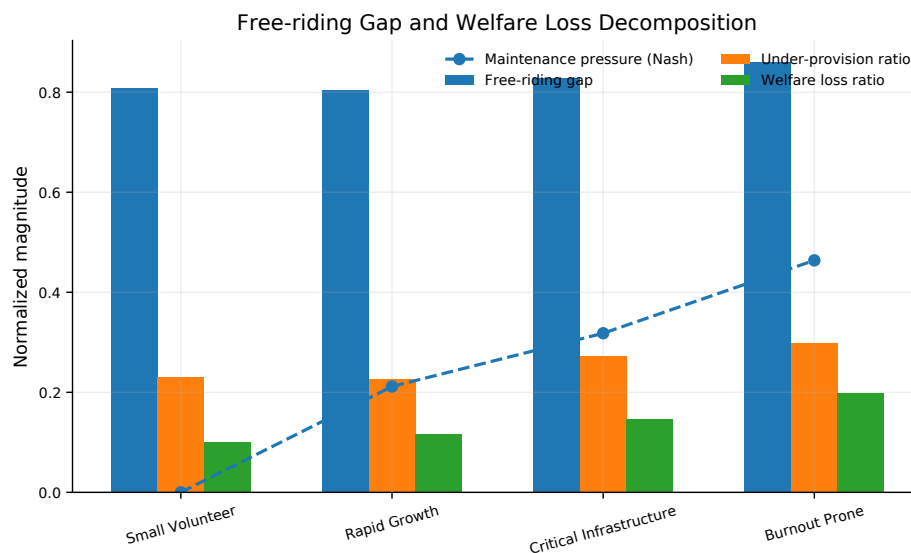


Figure 7: Welfare-loss decomposition under synthetic scenarios. The figure connects the free-riding gap, under-provision ratio, welfare loss ratio, and maintenance pressure index.

Figure 7 separates four related outcomes. The free-riding gap measures missing effective contribution, the under-provision ratio measures stock shortage, the welfare-loss ratio measures utility loss after benefit, cost, and pressure are combined, and maintenance pressure records the dynamic burden of the system. Reporting all four indicators prevents the analysis from collapsing free riding into a single number.

The decomposition also explains why scenario design matters. Burnout-prone provision has the most severe welfare loss because high cost suppresses effort and high pressure magnifies the damage from inadequate maintenance. Critical infrastructure has a smaller welfare-loss ratio, but its high pressure makes it a natural testbed for policy intervention. The simulation pattern is mechanism-oriented: free riding becomes more socially costly when contribution is expensive and pressure strongly harms stock.

7 Model IV: Policy Intervention Mechanisms

7.1 Policy Variables

We study six intervention mechanisms in addition to the baseline. The policy set is designed to cover both material incentives and behavioral incentives, reflecting the broader literature on cooperation, punishment, reputation, and warm-glow contribution.^[8, 9, 10]

- **Subsidy** reduces the private net cost of contribution.
- **Penalty** discourages high-benefit agents from contributing below a threshold.
- **Reputation** gives additional concave utility to contribution.
- **Matching fund** amplifies the effect of contribution on public-good stock.
- **Threshold governance** combines a minimum contribution threshold with budget and backlog reduction.
- **Combined portfolio** mixes subsidy, penalty, reputation, matching, budget, and backlog reduction.

Figure 8 maps the policy set. The incentive-side instruments mainly change private marginal returns, while the stock-pressure instruments affect the dynamic state or combine several channels.

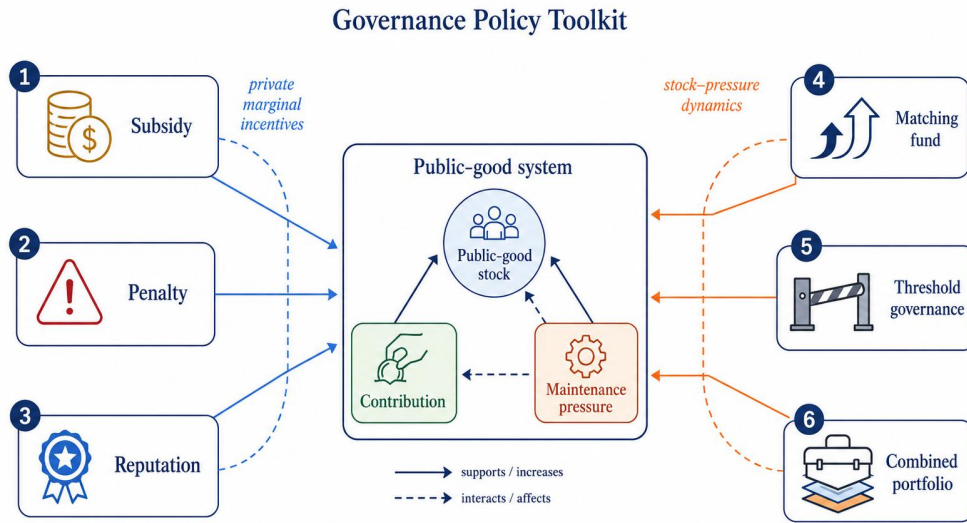


Figure 8: Governance policy toolkit considered in the DSPF framework. The diagram separates incentive-side instruments from stock-pressure instruments and shows how individual contribution, public-good stock, and maintenance pressure are connected in the policy layer.

Table 9 reports the policy intensities used in the deterministic comparison.

Table 9: Synthetic policy settings used in the deterministic policy comparison.

Policy	s	p	r	m	B	R	τ
Baseline	0.00	0.00	0.00	0.00	0.000	0.00	0.00
Subsidy	0.20	0.00	0.00	0.00	0.000	0.00	0.00
Penalty	0.00	0.35	0.00	0.00	0.000	0.01	0.16
Reputation	0.00	0.00	0.26	0.00	0.000	0.00	0.00
Matching fund	0.00	0.00	0.00	0.30	0.000	0.00	0.00
Threshold governance	0.00	0.18	0.00	0.00	0.025	0.05	0.18
Combined portfolio	0.12	0.14	0.16	0.18	0.020	0.04	0.14

These values are not empirical estimates; they are controlled settings chosen to compare pure and mixed intervention mechanisms under the same dynamic environment.

7.2 Policy Cost Function

Policy cost is explicitly included:

$$C_{\text{policy}} = s \sum_i e_i + 0.35m \sum_i e_i + 0.20r \log \left(1 + \sum_i e_i \right) + 50B + 35R + 0.03pN_F,$$

where N_F is the number of classified free riders. The coefficients are synthetic governance-cost weights. They keep administrative and intervention burdens from being ignored in the policy comparison. Table 10 gives their interpretation. These coefficients are relative accounting weights inside the simulation, not monetary prices; cost-effectiveness is therefore defined only under this stated accounting rule.

Table 10: Interpretation of the synthetic policy-cost coefficients.

Cost channel	Coefficient	Interpretation in the comparative policy experiment
Subsidy payment	1.00	Direct transfer proportional to contributed effort
Matching fund	0.35	Partial public matching cost for amplifying effective contribution
Reputation operation	0.20	Administrative and platform cost with concave scaling in total effort
Budget injection	50	Direct resource support to the public-good stock
Backlog reduction	35	Direct pressure-relief or maintenance operation cost
Penalty administration	0.03	Monitoring and enforcement cost per classified free-rider target

7.3 Marginal Effects of Policy Instruments

The policy terms change the individual marginal condition in different ways. For an interior effort level and away from the non-differentiable threshold point, subsidy contributes

$$\frac{\partial(se_i)}{\partial e_i} = s,$$

so it directly raises the private marginal return to effort. Reputation contributes

$$\frac{\partial r \log(1 + e_i)}{\partial e_i} = \frac{r}{1 + e_i},$$

which is positive but decreasing; it is therefore most effective for moving very low contributors away from zero. For an agent below the contribution threshold, the implemented penalty incentive is not a single quadratic term. It follows the same marginal-response rule used in the Nash-style update:

$$p (0.35 + 2.8(\tau^* - e_i)_+) \mathbf{1}_{\{b_i \geq \bar{b}_q\}} + 0.10p \mathbf{1}_{\{e_i < \tau^*\}}.$$

The first component targets high-benefit low contributors, while the second gives all below-threshold agents a small upward push. Matching fund changes the production side by multiplying $\partial G_{t+1}/\partial e_i$ by $(1 + m)$, so it increases the stock productivity of contribution rather than directly punishing free riding. Budget injection B shifts the stock equation, while backlog reduction R shifts the pressure equation. The six policies therefore intervene at different points of the marginal-incentive decomposition.

7.4 Multi-Criteria Evaluation

A policy is evaluated by a vector of indicators:

$$\mathcal{E}(\pi) = (\bar{W}, \bar{C}_{\text{policy}}, \Delta FR, \Delta H, \text{Gini}(e), \text{Stability}).$$

The tail averages are computed over the long-run part of each trajectory. In the numerical calculation this is the last $\max(8, \lfloor T/5 \rfloor)$ periods; under the reported simulation setting with $T = 30$, it is therefore the last eight periods. Effort inequality is measured by the Gini index

$$\text{Gini}(e) = \frac{2 \sum_{i=1}^N i e_{(i)}}{N \sum_{i=1}^N e_{(i)}} - \frac{N+1}{N},$$

where $e_{(i)}$ is the sorted effort vector. Local dynamic stability is summarized by

$$\text{Stability} = 1 - \rho(J_t),$$

where J_t is the local Jacobian of the state-transition map and $\rho(J_t)$ is its spectral radius. For this local stability diagnostic, the Jacobian is computed analytically from the unclipped saturating transition and interpreted as a local margin away from projection boundaries. Larger values are locally more favorable, while values close to zero should be read as near-neutral local margins rather than as empirical stability estimates. This multi-criteria design prevents policy choice from being reduced to welfare alone; cost, pressure, fairness, and stability are also governance constraints.

For the normalized decision matrix, each criterion is scaled within the displayed policy set. For a benefit-type criterion x , we use

$$S_x(\pi) = \frac{x(\pi) - \min_{\pi'} x(\pi')}{\max_{\pi'} x(\pi') - \min_{\pi'} x(\pi')},$$

and for a cost-type criterion such as policy cost or effort inequality, the numerator is reversed. If a column has zero range, it is treated as a neutral column in the visualization. The normalization is used only for visual comparison; the tables retain the original numerical values.

8 Model Verification and Numerical Reliability

Before interpreting policy outcomes, we check the internal consistency of the numerical framework. Table 11 summarizes the reliability checks used in the study. These checks do not convert synthetic scenarios into empirical evidence; they show that the conclusions follow from a reproducible computational design. Numerical values are rounded to displayed precision, so independent reruns may differ at machine precision while preserving the reported rankings and qualitative conclusions.

Table 11: Model checking and numerical reliability evidence.

Check	Evidence	Interpretation
Reproducible synthetic route	Fixed seed 20260614; four scenarios; seven policies; all inputs and outputs are treated as controlled synthetic simulations.	Results can be regenerated and interpreted as controlled simulations.
Parameter audit	Scenario parameters, behavioral weights, solver settings, policy intensities, and policy-cost weights are separated in Tables 3, 5, 9, and 10.	Assumptions, calibration choices, and outputs are not mixed.
Best-response convergence	The damped update reaches the stopping tolerance in 27–28 sweeps before the 60-sweep limit.	The individual-rational benchmark is a converged numerical fixed point of the implemented response map, while still being treated as a Nash-style proxy rather than a unique analytical equilibrium.
Planner benchmark validity	The stage-wise L-BFGS-B solution gives higher contribution and welfare than the decentralized benchmark in all four scenarios.	The planner benchmark is a consistent numerical upper reference for diagnosing under-provision; global analytical optimality is not claimed.
Scenario separation	Small volunteer, rapid growth, critical infrastructure, and burnout-prone settings stress different cost, pressure, and demand mechanisms.	The scenario analysis is not a repetition of one parameter setting.
Uncertainty tests	One-dimensional sensitivity, two-dimensional and fine-grid three-dimensional pure subsidy–penalty response surfaces, perturbed-scenario robustness, joint behavioral-response perturbation, one-at-a-time weight audit, solver-damping checks, Monte Carlo runs, and Pareto search are all reported.	The qualitative conclusions are tested beyond one baseline parameter vector and beyond one marginal-response calibration.
Displayed-precision reproducibility	The numerical design records the fixed random seed and synthetic settings; optimization results are interpreted at table-rounding precision.	Small floating-point differences do not change the reported ranking or qualitative conclusion.

9 Numerical Experiments and Results

9.1 Baseline Dynamic Behavior

Figure 9 summarizes stock, pressure, free-riding, and welfare trajectories under baseline governance. It is placed in the results section because it functions as numerical evidence from the synthetic simulation rather than as part of the model definition.

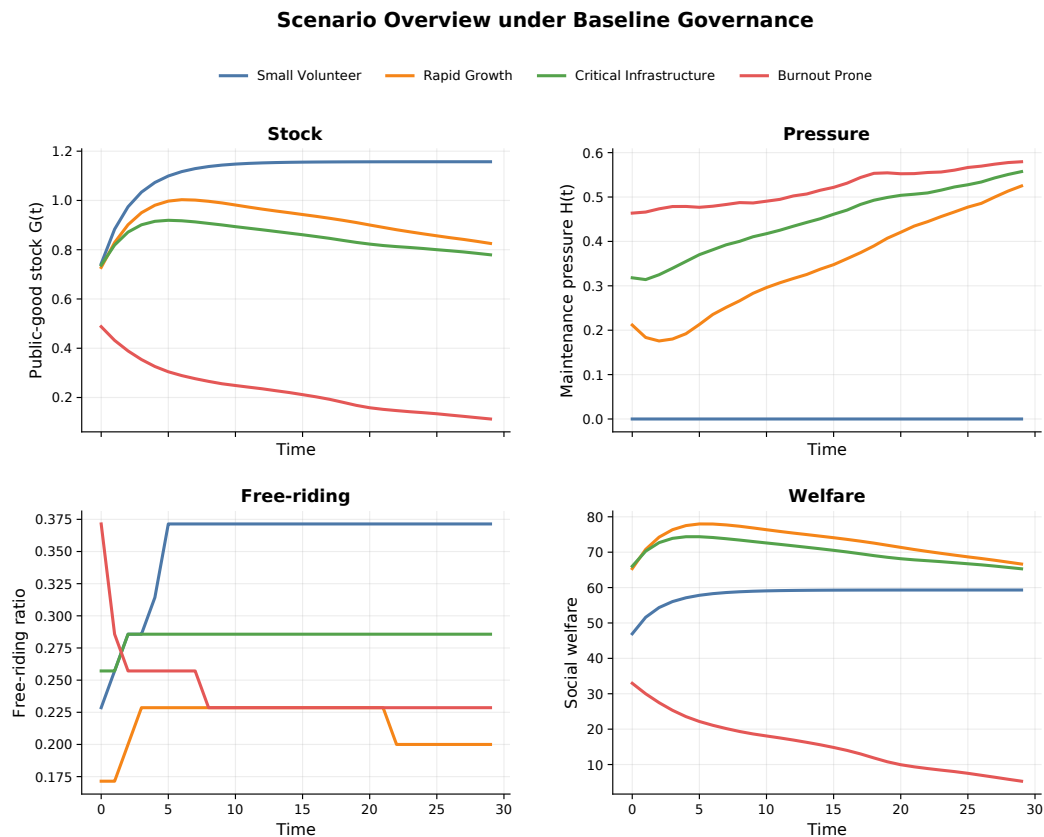


Figure 9: Baseline dynamic behavior under synthetic scenarios. The panels compare stock, maintenance pressure, free-riding ratio, and welfare under the baseline Nash-style simulation.

The upper panels of Figure 9 show that stock and maintenance pressure are coupled but not identical. A scenario may keep a moderate stock for several periods while pressure continues to build. The lower panels show that free riding and welfare also vary by scenario. Free riding is therefore not only an individual-level behavior; through stock and pressure feedback, it becomes a dynamic system-level problem.

The baseline simulations indicate that free riding and under-provision are not uniform across scenarios. Small-volunteer provision reaches a relatively stable stock with almost no maintenance pressure, but still exhibits a nonzero free-riding ratio. This means that a system can appear operational even when the contribution burden is unevenly distributed. Rapid growth and critical infrastructure display higher maintenance pressure, showing that demand growth and infrastructure importance make the public-good problem more dynamic. The burnout-prone scenario shows the most severe welfare loss when compared with the social-planner benchmark.

The baseline results serve two purposes. First, they show that the synthetic scenarios are not redundant: each stresses a different part of the mechanism. Second, they justify using the critical-infrastructure scenario for the detailed policy comparison. It is neither trivially stable nor fully collapsed, and it contains enough free riding, pressure, and welfare loss for policy differences to be visible. The results remain synthetic numerical patterns, not real-world measurements.

Figure 10 gives a three-dimensional phase view of representative critical-infrastructure trajectories in stock–pressure–welfare space. It is a geometric summary of the same scenario used in the policy and robustness sections. The baseline path remains in a lower-welfare, higher-pressure region, whereas reputation and combined portfolio policies move the trajectory toward higher stock and welfare. The separation of the paths indicates that policy effects operate through the coupled stock–pressure system, not only through an immediate change in effort.

The phase-space view is useful because two trajectories may have similar short-run welfare but different pressure accumulation. A high-pressure path is more fragile: after a demand or decay shock, the same contribution level may no longer maintain stock. The trajectory figure therefore links the baseline dashboard to the policy section and explains why the critical-infrastructure scenario is evaluated by both welfare and dynamic resilience.

The main modeling message is that public-good failure is not merely a low-contribution event. It is a state-space drift: weak private incentives reduce contribution, insufficient contribution raises pressure, and pressure then weakens future welfare. The figure is placed before the policy-comparison table so that the dynamic failure channel is visible before the interventions are evaluated.

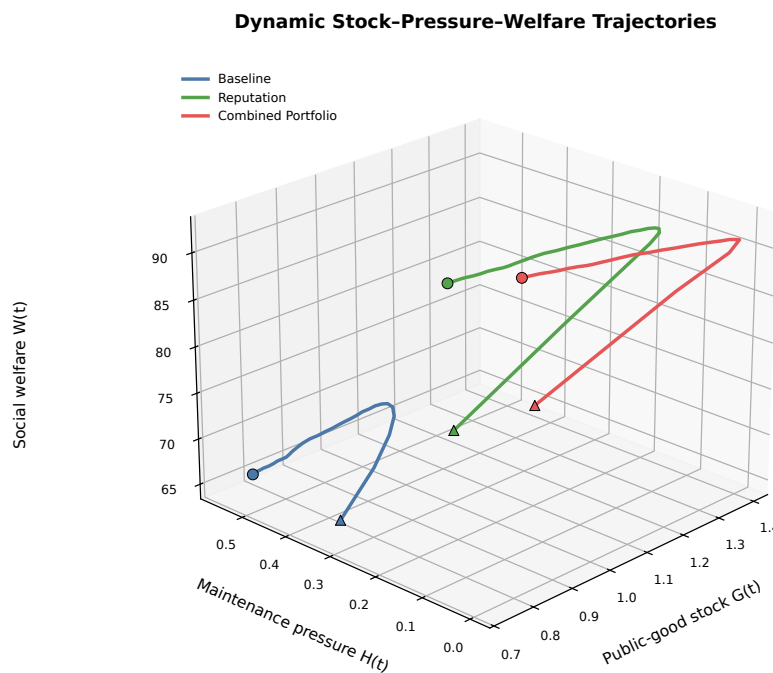


Figure 10: Three-dimensional phase trajectory in the stock-pressure-welfare space for the critical-infrastructure synthetic scenario. The figure visualizes how baseline governance, reputation, and combined portfolio policies move through different dynamic regions.

The phase trajectory also explains why welfare alone is insufficient. The baseline path first reaches a relatively high stock, then drifts toward higher pressure and lower welfare, so short-run service quality can conceal accumulated maintenance burden. Reputation shifts the trajectory upward at limited cost by changing marginal incentives. The combined portfolio follows a different path: it does not simply raise instantaneous welfare, but pulls the system toward a lower-pressure region. For public-good governance, a policy with slightly lower welfare may still be preferable if it reduces the risk of a future pressure-driven decline.

9.2 Policy Comparison in the Critical-Infrastructure Scenario

The critical-infrastructure scenario is selected for the main policy comparison because it combines high public-good importance, visible free riding, and nontrivial maintenance pressure. Table 12 and Figure 11 show that subsidy, penalty, reputation, threshold governance, and the combined portfolio all improve welfare relative to the baseline. Under the updated capacity-saturating dynamics, the combined portfolio is marginally highest in long-run welfare

and strongest in maintenance-pressure reduction, but it also has the highest policy cost. Reputation produces nearly the same welfare level with a much smaller cost, so it is the strongest low-cost behavioral instrument.

Table 12: Policy comparison in the critical-infrastructure synthetic scenario.

Policy	Welfare	Cost	Free riding	Pressure	ΔFR	ΔH	Effort Gini	Stability
Baseline	66.52	0.00	0.286	0.533	0.000	0.000	0.451	0.003
Subsidy	82.16	1.90	0.057	0.388	0.229	0.145	0.412	-0.004
Penalty	76.59	0.35	0.000	0.469	0.286	0.063	0.392	-0.001
Reputation	83.74	0.12	0.000	0.392	0.286	0.141	0.358	-0.004
Matching fund	72.77	0.27	0.286	0.549	0.000	-0.017	0.450	-0.000
Threshold governance	75.62	3.02	0.089	0.443	0.196	0.090	0.399	-0.001
Combined portfolio	83.81	4.59	0.000	0.311	0.286	0.222	0.360	-0.006

Tail averages over the last eight periods are used for long-run indicators. ΔFR and ΔH are reductions relative to the baseline; a negative ΔH means pressure is higher than baseline. Stability is the local margin $1 - \rho(J_t)$ from the unclipped transition and is used as a relative diagnostic only.

Unlike Table 8, Table 12 uses long-run tail averages over the last eight periods. The comparison therefore measures sustained policy performance rather than a one-period response gap. This distinction explains why the baseline welfare level in the policy table is not identical to the one-step individual-rational benchmark. The stability margins are all close to zero, so the stability column is used only to distinguish relative local behavior, not to declare empirical dynamic stability or instability.

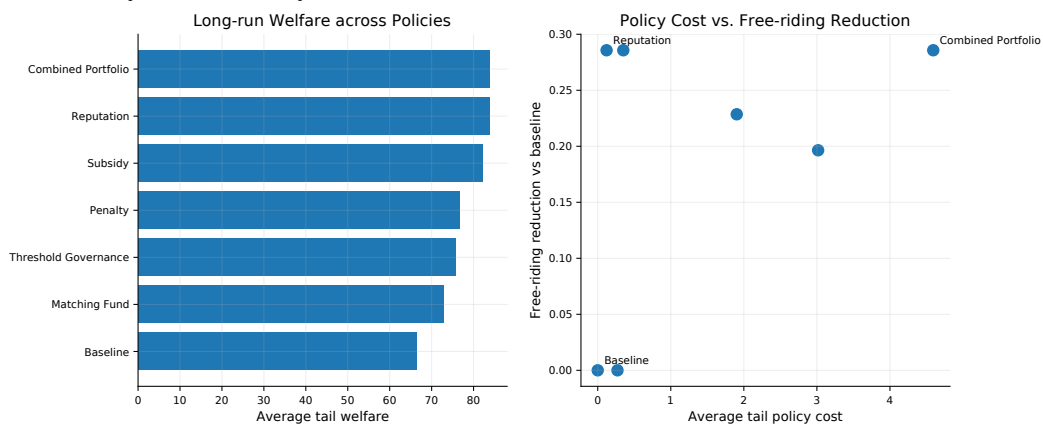


Figure 11: Policy comparison under the critical-infrastructure synthetic scenario. Interventions improve welfare and reduce free riding, but differ in policy cost and pressure reduction.

The comparison gives three main patterns. First, penalty, reputation, and the combined portfolio eliminate classified free riding in the critical-infrastructure setting. Penalty changes the downside of under-contribution for high-benefit agents, whereas reputation creates a positive private return to effort. Second, the matching fund improves welfare relative to the baseline but does not reduce the classified free-riding ratio or relieve pressure under the current parameterization. It raises the productivity of existing contribution without directly changing the high-benefit low-effort classification. Third, the combined portfolio gives the strongest pressure reduction but also the highest policy cost because it changes both incentives and state dynamics. The preferred policy therefore depends on the objective: reputation is attractive for cost-effective incentive correction, while the combined portfolio is more attractive when pressure is the dominant risk.

The policy comparison also explains why we avoid a single “optimal policy” claim. The combined portfolio is only slightly above reputation in welfare but gives larger pressure

reduction; reputation has much lower cost; penalty provides strong low-cost free-riding correction. These rankings differ across criteria. For public-good governance, incentive correction, administrative cost, fairness, and system stability are rarely optimized by one instrument. The resulting rule is conditional: use reputation when low-cost incentive correction is sufficient, and use a combined portfolio when pressure relief is part of the objective.

9.3 Multi-Dimensional Policy Ranking

Figure 12 compares selected policies through a normalized multi-criteria decision matrix. Welfare, free-riding reduction, pressure relief, fairness, and local stability are scaled so that larger values are preferred, while policy cost and effort inequality are reversed before scaling. The raw stability margins in Table 12 are close to zero; their normalized column is therefore a relative local diagnostic, not an empirical stability claim. The figure is a compact decision table rather than a decorative summary.

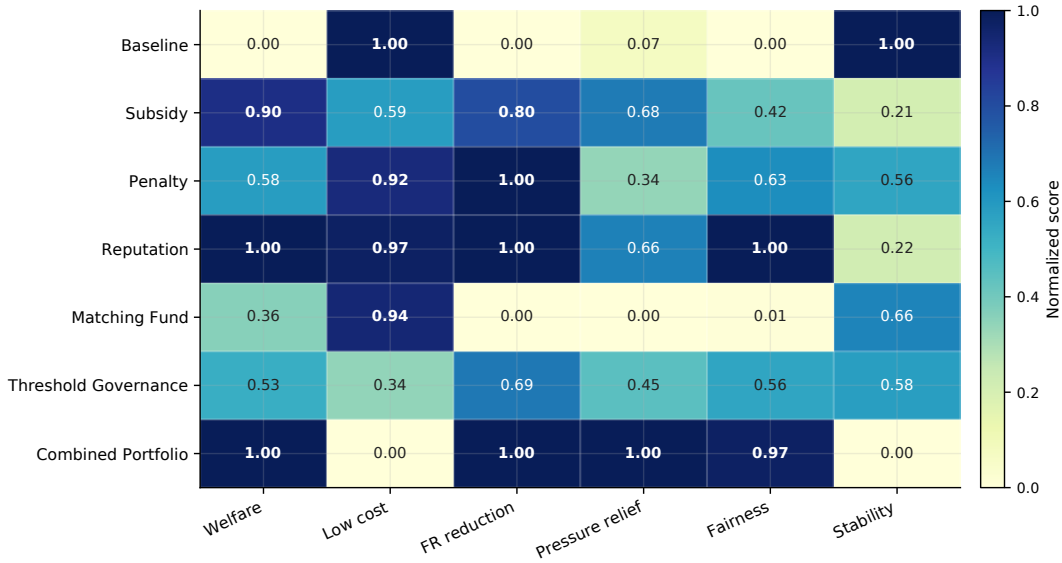


Figure 12: Normalized policy decision matrix under the critical-infrastructure scenario. Each column is normalized within the displayed policy set, and a larger value indicates a more desirable performance on that criterion.

The matrix shows why no policy dominates in all dimensions. The combined portfolio is strong in welfare, free-riding reduction, pressure relief, and fairness, but weak in cost and lower in stability under this normalization. Reputation performs well on welfare, free-riding reduction, and fairness with much lower cost, while penalty is strong for free-riding reduction but less balanced in pressure relief. The evidence therefore favors a policy-selection rule rather than a universal maximizer: low-pressure settings can use lighter incentives, whereas infrastructure-like settings may justify a more expensive portfolio when pressure relief is decisive.

This visual form is useful because the model contains both agent-level and system-level indicators. Free-riding reduction measures behavior, maintenance pressure measures system burden, and welfare combines benefit and cost. A recommendation based on only one indicator would ignore part of the public-good problem.

9.4 Pareto Trade-Off Analysis

Definition 1 (Policy dominance). For two policy portfolios π_a and π_b , policy π_a dominates π_b if it is no worse in all six implemented indicators,

$$\begin{aligned} \overline{W}(\pi_a) &\geq \overline{W}(\pi_b), & \overline{G}(\pi_a) &\geq \overline{G}(\pi_b), & \overline{C}_{\text{policy}}(\pi_a) &\leq \overline{C}_{\text{policy}}(\pi_b), \\ \overline{FR}(\pi_a) &\leq \overline{FR}(\pi_b), & \overline{H}(\pi_a) &\leq \overline{H}(\pi_b), & \text{Gini}(\pi_a) &\leq \text{Gini}(\pi_b), \end{aligned}$$

and at least one of these inequalities is strict. A policy is Pareto-efficient if no sampled policy dominates it.

We use a random policy search to examine trade-offs among welfare, stock, cost, free-riding reduction, pressure, and effort inequality. Figure 13 shows a two-dimensional projection of the six-indicator dominance test: a policy is dominated when another sampled portfolio is no worse in all six indicators and strictly better in at least one. The efficient points usually combine modest penalty, reputation, and selective backlog reduction. The key message is not a single optimal vector, but the existence of a policy frontier.

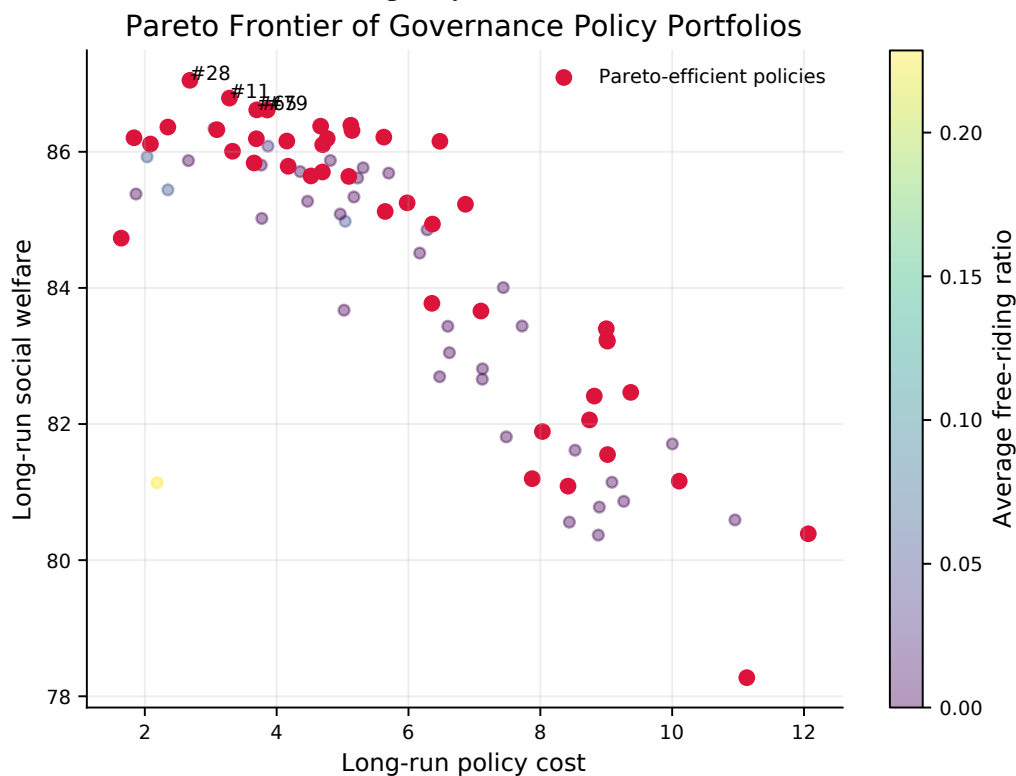


Figure 13: Pareto frontier for synthetic policy trade-offs. The plot shows welfare against policy cost, with color indicating free-riding ratio; highlighted points are Pareto-efficient under the full six-indicator dominance rule used in the numerical calculation.

Figure 13 provides the most direct trade-off view, but it should be read as a projection of the full dominance calculation rather than as a purely two-dimensional frontier. A dominated point means that another sampled policy portfolio achieves at least as much welfare and stock, no greater cost, no worse free-riding, no higher pressure, and no higher effort inequality, with at least one strict improvement. The frontier points represent policies that remain competitive after these comparisons. For policy design, this is more informative than reporting a single maximum-welfare setting, because real decision makers often face budget limits and institutional constraints.

Figure 14 adds a three-dimensional projection of the same sampled policy search. The vertical and depth coordinates help reveal a pattern that can be hidden in the two-dimensional cost-welfare plot: some portfolios with similar welfare and cost can still differ in pressure control and effort inequality. Therefore, the Pareto analysis is interpreted as a multi-criteria selection problem rather than as a one-dimensional ranking.

The scatter pattern explains why we do not report one parameter vector as “the optimum”. With limited governance capacity, a decision maker may prefer slightly lower welfare in exchange for lower cost and lower inequality. If reliability is the priority, a higher-cost point

may be justified by lower pressure. The sampled frontier is therefore used as a decision map, not a deterministic prescription.

The Pareto view also guards against a common reporting error: selecting the largest welfare value while ignoring side effects. In the sampled set, some high-welfare portfolios become dominated once cost, free riding, pressure, stock, and effort inequality are considered together. Policy should therefore be selected by trade-off position rather than by a single scalar score.

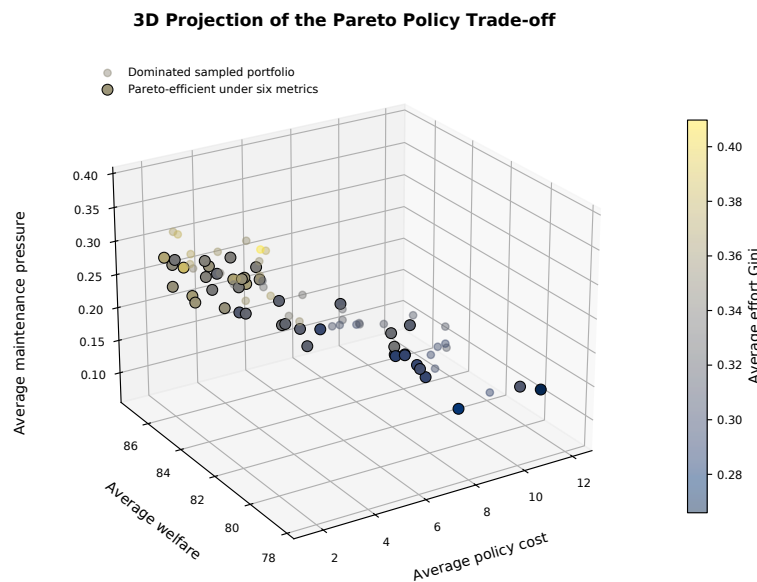


Figure 14: Three-dimensional projection of the synthetic Pareto policy search. Highlighted portfolios remain Pareto-efficient under the full six-indicator dominance rule.

The three-dimensional projection shows that efficient points are not concentrated at the lowest-cost edge or at the maximum-welfare corner. Several efficient portfolios lie in a middle region with moderate cost, high welfare, and acceptable effort inequality. Once free riding is largely corrected, additional spending may produce smaller welfare gains and may worsen the fairness–cost balance. The frontier should therefore be read as a screening tool for feasible portfolios, not as an instruction to choose the most expensive or most punitive setting.

The Pareto analysis also prevents the recommendation from being tied too closely to one named policy. Reputation and penalty are interpretable mechanisms, but actual governance may combine them at different intensities. The frontier experiment shows that mixed portfolios can be competitive, especially when incentive correction is paired with moderate pressure relief. For this reason, the recommendation is stated as principles and scenario-specific portfolios rather than exact numerical rates.

10 Sensitivity, Robustness, and Uncertainty Analysis

10.1 One-Dimensional Sensitivity

We vary key model and policy parameters in the critical-infrastructure scenario to test whether the conclusions depend on one parameter setting. Table 13 lists the parameters with the largest welfare ranges. Benefit scale is the dominant welfare driver, followed by contribution-to-stock conversion, reputation, demand-to-pressure conversion, and subsidy.

Reputation, subsidy, and penalty can reduce classified free riding almost to zero, but their pressure effects differ. Incentive correction is therefore necessary but not sufficient; pressure control also depends on the stock–pressure channel.

Table 13: One-dimensional sensitivity summary in the critical-infrastructure synthetic scenario.

Parameter	Tested range	Welfare range	Free-riding range	Pressure range
benefit_scale	[0.810, 1.890]	35.90–102.16	0.257–0.286	0.478–0.503
alpha	[0.468, 1.092]	48.82–76.01	0.282–0.286	0.444–0.510
reputation	[0.000, 0.600]	68.63–86.86	0.000–0.286	0.253–0.489
lambda	[0.264, 0.616]	61.76–78.00	0.257–0.286	0.301–0.613
subsidy	[0.000, 0.600]	68.63–83.80	0.000–0.286	0.203–0.489
eta	[0.180, 0.420]	62.59–74.56	0.257–0.286	0.460–0.522
penalty	[0.000, 0.600]	68.63–79.83	0.000–0.286	0.419–0.489

10.2 Subsidy-Penalty Response Surface

We conduct a pure subsidy–penalty grid experiment in the critical-infrastructure scenario. The compact two-dimensional heatmap uses a 9×9 grid for tabular sensitivity reporting, while the three-dimensional response surface uses a separate fine grid with $13 \times 13 = 169$ tested policy combinations. In both grids, only subsidy s and penalty p are varied; threshold governance, backlog reduction, matching support, reputation, and direct budget injection are kept at their baseline values. Thus, the grid origin $(s, p) = (0, 0)$ is exactly the no-policy baseline, and the surface isolates the two incentive-side instruments rather than mixing them with pressure-relief tools.

On the fine grid, the highest-welfare tested point is subsidy 0.150 and penalty 0.600, with average tail welfare 82.715 and zero classified free riding. This point lies on the penalty boundary of the tested grid, so it should be read as the best point among the sampled combinations rather than as an interior optimum or a global policy prescription. The leading region combines moderate subsidy with medium-to-strong penalty. In this calibrated model, reward and deterrence are complementary: subsidy raises broad contribution incentives, while penalty targets high-benefit low-effort behavior. Since the scan uses only two levers, its welfare gain is smaller than that of portfolios with direct pressure relief, but the comparison more cleanly identifies the marginal incentive mechanism.

Table 14: Top welfare points in the fine-grid pure subsidy-penalty response-surface experiment.

Subsidy	Penalty	Welfare	Free riding	Pressure	Policy cost	Effort Gini
0.150	0.600	82.715	0.000	0.376	1.581	0.393
0.150	0.550	82.669	0.000	0.379	1.548	0.394
0.150	0.500	82.615	0.000	0.382	1.515	0.396
0.200	0.500	82.575	0.000	0.359	2.371	0.375
0.200	0.450	82.573	0.000	0.362	2.327	0.377

Figure 15 visualizes the fine pure subsidy–penalty grid as a response surface. Welfare is the height of the surface, and color represents average maintenance pressure. The red marker denotes the highest-welfare tested grid point, located on the penalty boundary of the scan. The response surface should not be read as a recommendation to maximize punishment or subsidy mechanically; it is a controlled sensitivity experiment within the implemented cost and behavioral-response assumptions.

The grid also clarifies the asymmetric roles of subsidy and penalty. Penalty reduces classified free riding more directly, while subsidy prevents the response from relying only on punishment. Because the scan excludes backlog reduction and direct budget support, however,

the pressure field remains above the combined-portfolio pressure level. Incentive correction improves welfare and free-riding behavior, but it does not fully replace pressure-relief or capacity-oriented instruments when maintenance burden is the main risk.

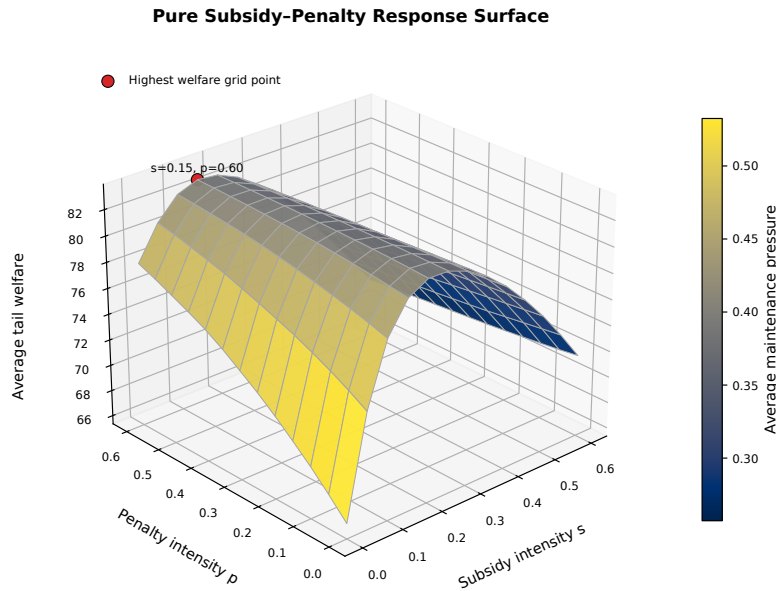


Figure 15: Three-dimensional pure subsidy-penalty response surface on a 13×13 fine grid in the critical-infrastructure synthetic scenario. Welfare is the vertical axis, and color represents average maintenance pressure. The grid varies only subsidy and penalty, so the origin equals the no-policy baseline. The red marker denotes the highest-welfare tested grid point, which lies on the penalty boundary of the scan.

10.3 Robustness under Perturbed Scenarios

For robustness, we perturb decay, cost, and demand in the critical-infrastructure scenario while keeping the agent population, policy definitions, behavioral-response weights, and evaluation window fixed. Table 15 compares baseline and combined-portfolio outcomes under this stress test. In every tested perturbation, the combined portfolio improves welfare and reduces classified free riding to zero. The demand-shock case still has high pressure after intervention, but its pressure is below the corresponding baseline. Thus the portfolio provides behavioral correction and partial pressure relief, while structural demand growth still calls for capacity-oriented management.

Table 15: Robustness comparison between baseline and combined portfolio under perturbed critical-infrastructure synthetic settings.

Variant	Welfare		Free-riding ratio		Pressure	
	Baseline	Combined	Baseline	Combined	Baseline	Combined
Demand shock	57.72	78.15	0.257	0.000	0.684	0.531
High cost	66.34	84.58	0.286	0.000	0.505	0.286
High decay	67.00	84.25	0.286	0.000	0.492	0.260
Low cost	70.38	85.29	0.282	0.000	0.491	0.239
Low decay	69.56	85.66	0.286	0.000	0.507	0.267

Note. Each row perturbs only the named stress factor around the critical-infrastructure scenario; the policy definitions and behavioral calibration remain fixed.

10.4 Behavioral-Response and Solver-Damping Robustness

The Nash-style marginal-response rule contains five behavioral-response weights that scale perceived stock benefit, perceived pressure relief, targeted penalty pressure, threshold-gap curvature, and the mild threshold push. These constants are baseline calibration weights rather than empirical estimates. We therefore run two complementary checks in the critical-infrastructure scenario. First, all five weights are multiplied by a common factor $\gamma \in \{0.8, 1.0, 1.2\}$ while the social-planner problem is left unchanged. Second, each weight is perturbed one at a time by multipliers 0.70, 0.85, 1.00, 1.15, 1.30 while the other four weights remain fixed at their baseline values.

Table 16 shows that the benchmark conclusion is stable under joint scaling. For all three values of γ , Nash-style individual-rational contribution remains far below the stage-wise planner benchmark. Larger behavioral responsiveness raises Q^{IR} and lowers the welfare-loss ratio, but under-provision remains. The free-riding gap is therefore not produced by one arbitrary behavioral-weight setting.

Table 16: Joint behavioral-response weight robustness in the critical-infrastructure synthetic scenario. All five Nash-style behavioral weights are multiplied by γ .

γ	Q^{IR}	Q^{SO}	Under-provision	Welfare loss	Free-riding	Pressure ^{IR}
0.8	2.645	17.453	0.289	0.160	0.257	0.328
1.0	3.007	17.453	0.272	0.147	0.257	0.318
1.2	3.335	17.453	0.258	0.136	0.229	0.310

The joint-scaling test is intentionally asymmetric: it perturbs the decentralized response rule but leaves the planner benchmark fixed. It asks whether the under-provision conclusion disappears when private agents react more strongly to stock benefit and pressure relief. Over the tested range, it does not; higher responsiveness narrows the gap, but the social benchmark remains substantially larger.

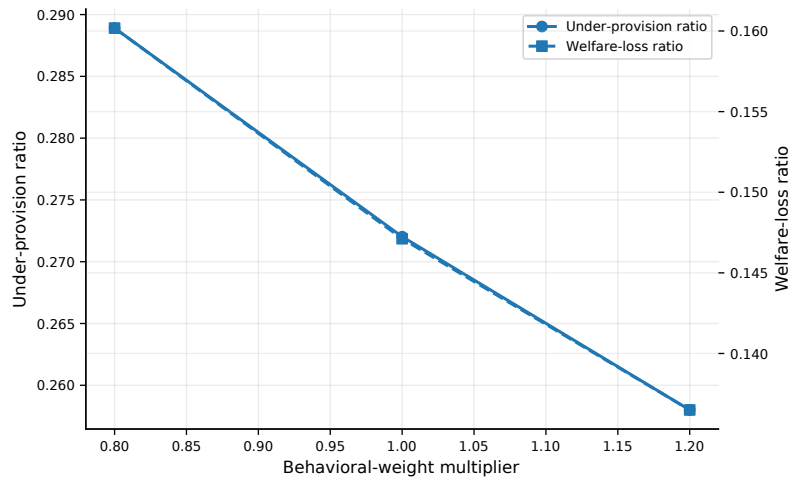


Figure 16: Joint behavioral-weight robustness of under-provision and welfare-loss ratios in the critical-infrastructure synthetic scenario. Increasing behavioral responsiveness lowers both ratios, but the under-provision conclusion remains unchanged.

The one-at-a-time audit in Table 17 gives a sharper interpretation of the calibration weights. The baseline conclusion is most sensitive to the perceived stock-benefit response θ_G , mildly sensitive to perceived pressure relief θ_H , and unchanged by the three penalty-related weights in the no-policy baseline. This is expected: when $p = 0$, the penalty-base, penalty-gap, and threshold-push terms are inactive. They become relevant only in policy-response experiments with penalty or threshold governance.

Table 17: One-at-a-time behavioral-weight robustness in the critical-infrastructure baseline. Each row reports the range generated by multipliers 0.70, 0.85, 1.00, 1.15, 1.30 applied to one weight while all other weights remain fixed.

Perturbed weight	Q^{IR} range	Under-provision range	Welfare-loss range	Free-riding range
Stock-benefit response θ_G	2.516–3.437	0.254–0.295	0.133–0.165	0.229–0.286
Pressure-relief response θ_H	2.947–3.066	0.269–0.275	0.145–0.149	0.257–0.257
Penalty base ξ_0	3.007–3.007	0.272–0.272	0.147–0.147	0.257–0.257
Penalty gap ξ_1	3.007–3.007	0.272–0.272	0.147–0.147	0.257–0.257
Threshold push ξ_L	3.007–3.007	0.272–0.272	0.147–0.147	0.257–0.257

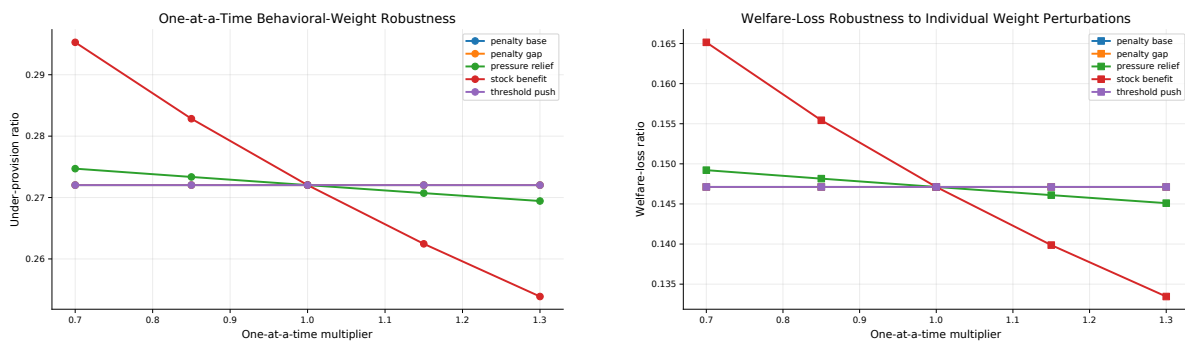


Figure 17: One-at-a-time behavioral-weight robustness. The stock-benefit response has the clearest effect on under-provision and welfare loss, while penalty-related weights are inactive in the no-policy baseline because $p = 0$.

The behavioral perturbations also preserve the representative ordering of baseline, reputation, and combined-portfolio governance in the robustness rerun. Baseline remains dominated; reputation remains the highest-welfare low-cost option; and the combined portfolio stays close in welfare while providing stronger stock and pressure support at a higher cost. The

policy interpretation therefore does not rest on the baseline response weights alone.

We also test sensitivity to the damping coefficient in the marginal-response solver. The raw best-response weight is changed from 0.42 to 0.30 and 0.55, with the maximum sweep limit fixed at 60. Table 18 shows convergence in all three settings, and aggregate contribution, under-provision, and welfare loss are unchanged at reported precision. Larger damping reduces the number of sweeps, but the substantive benchmark is stable. The Nash-style proxy is therefore not driven by the baseline damping coefficient.

Table 18: Solver-damping robustness for the Nash-style marginal-response update.

Raw weight	Sweeps	Converged	Raw residual	Damped residual	Q^{IR}	Welfare loss
0.30	41	Yes	1.13×10^{-14}	3.39×10^{-15}	3.007	0.147
0.42	27	Yes	$< 10^{-16}$	2.78×10^{-17}	3.007	0.147
0.55	19	Yes	$< 10^{-16}$	2.78×10^{-17}	3.007	0.147

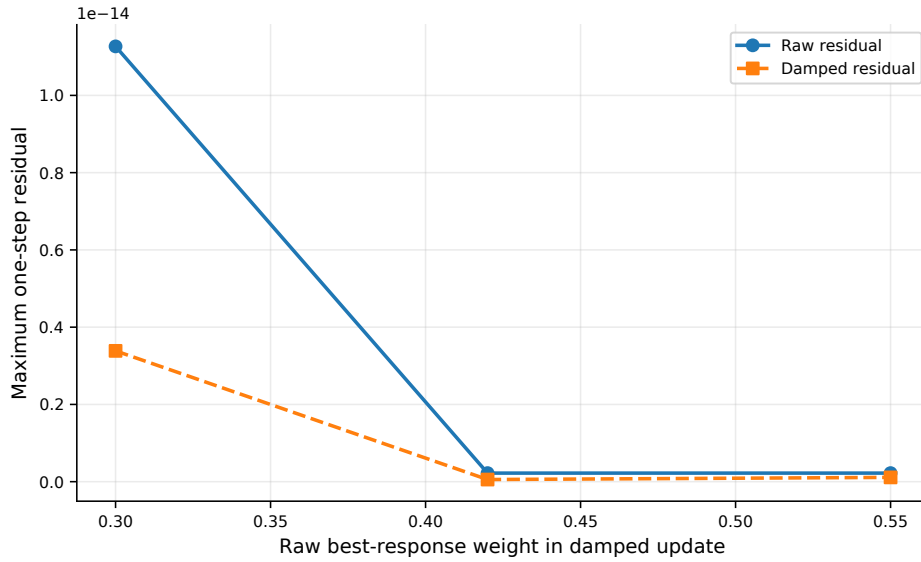


Figure 18: Residual behavior under alternative damping coefficients for the Nash-style marginal-response solver. All three damping settings converge within the sweep limit, and the aggregate contribution and welfare-loss ratios in Table 18 remain stable.

10.5 Monte Carlo Uncertainty Analysis

Monte Carlo experiments perturb parameters and rerun the critical-infrastructure scenario under baseline, reputation, and combined-portfolio policies. Table 19 and Figure 19 show that both interventions shift the welfare distribution upward and sharply reduce free riding. Under the updated capacity-saturating dynamics, the combined portfolio has the highest mean welfare and almost zero classified free riding, while reputation delivers nearly the same mean welfare at a much lower average policy cost. The uncertainty analysis therefore supports a conditional recommendation: use reputation when the priority is low-cost incentive correction, and use the combined portfolio when pressure control and system protection justify higher intervention cost.

Table 19: Monte Carlo uncertainty summary under synthetic perturbations.

Policy	Welfare mean	Welfare std	Final G mean	Free-riding mean	Cost mean
Baseline	76.53	15.82	0.830	0.253	0.00
Combined portfolio	93.92	13.68	1.287	0.003	4.60
Reputation	93.54	13.70	1.172	0.017	0.12

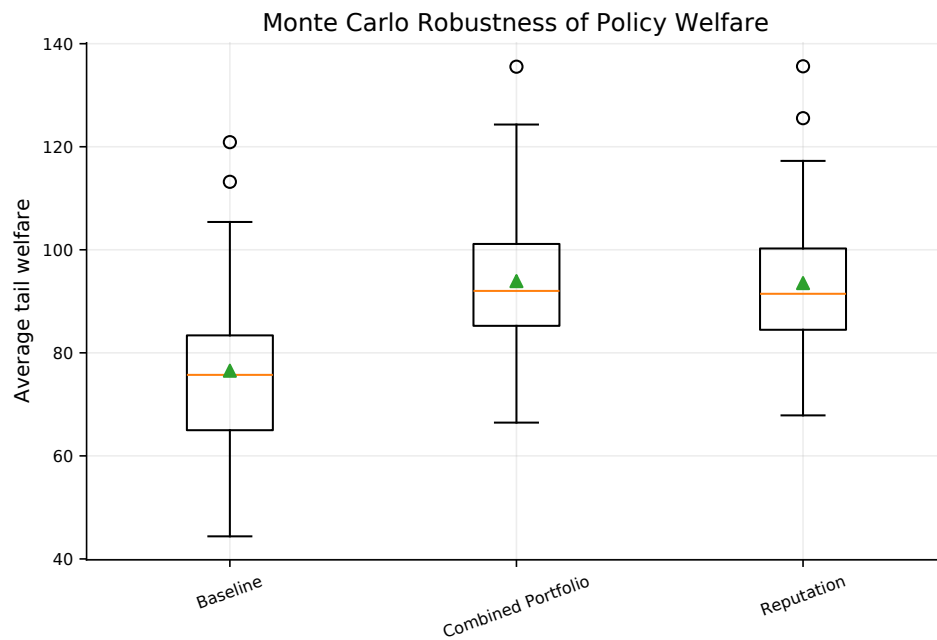


Figure 19: Welfare distribution under Monte Carlo synthetic perturbations. The boxplot shows that reputation and combined portfolio policies shift the welfare distribution upward relative to the baseline; the corresponding free-riding means are reported in Table 19.

Figure 19 complements the deterministic policy comparison. The deterministic table reports one controlled setting, whereas Monte Carlo runs show outcome variation after parameter perturbation. Baseline welfare has a wide spread, indicating sensitivity of the uncontrolled system to parameter uncertainty. Reputation and combined portfolio policies shift the welfare distribution upward and reduce free riding, so the intervention conclusion is not tied to one exact parameter vector.

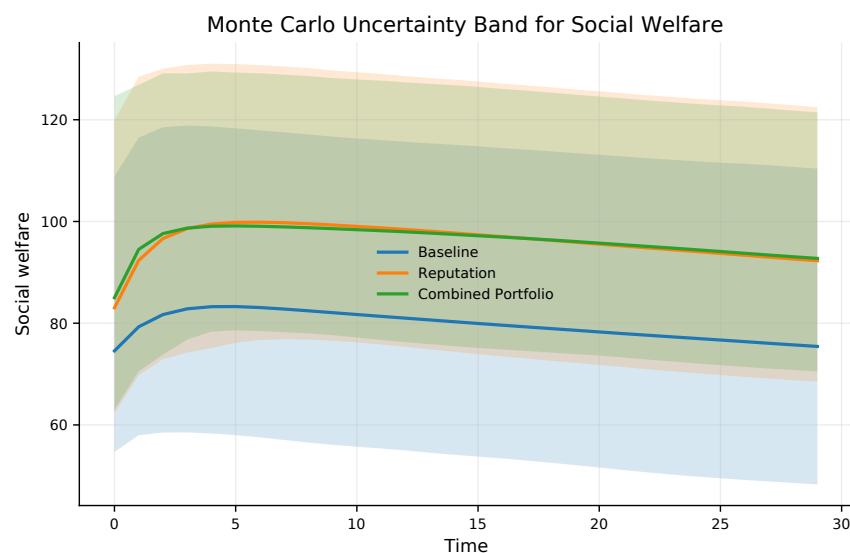


Figure 20: Time-varying welfare uncertainty bands under Monte Carlo synthetic perturbations. Lines show mean welfare paths and shaded bands show simulation uncertainty.

Figure 20 adds the time dimension. The baseline path remains more volatile across the simulated horizon, whereas reputation and combined portfolio policies keep welfare in a higher and narrower region after the initial adjustment. The Monte Carlo result is therefore interpreted as both a level improvement and a robustness improvement.

The two improved policies play different roles. Reputation achieves nearly the same mean

welfare as the combined portfolio at very low cost, making it a cost-effective behavioral instrument. The combined portfolio has the highest mean welfare and higher cost, but almost eliminates free riding and gives stronger pressure-oriented protection. The uncertainty analysis therefore supports a conditional recommendation: use reputation for low-cost incentive correction, and use combined portfolios when maintenance pressure or public-good importance is high.

11 Policy Recommendations

The numerical experiments indicate that governance should be selected by scenario condition rather than by a universal instrument. The following rules translate the model outputs into operational recommendations. They are mechanism-based decision rules under the synthetic DSPF framework, not empirical prescriptions for a specific public-good system. Figure 21 summarizes the scenario-based logic.

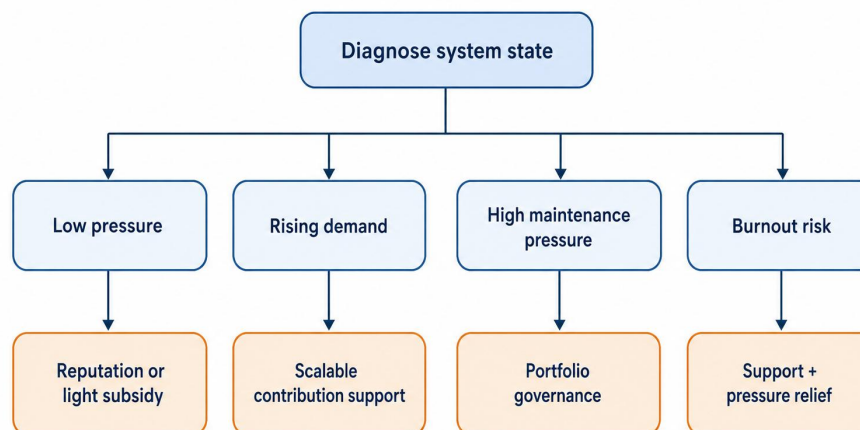


Figure 21: Scenario-based governance decision tree derived from the synthetic DSPF experiments. The decision logic maps diagnosed system states to recommended governance responses; it does not report empirical thresholds or numerical values.

11.1 Recommendation 1: Use Low-Cost Incentive Correction Before Heavy Intervention

The individual-rational and social-planner comparison shows systematic under-provision under the synthetic settings. The first governance step should therefore correct private marginal incentives. Reputation mechanisms, targeted penalties, and subsidies can raise effort by increasing private marginal benefit or lowering effective private cost.

For low-pressure voluntary settings, the recommended decision rule is: start with reputation or light subsidy, monitor the free-riding ratio and effort inequality, and avoid large direct budget injections unless maintenance pressure begins to accumulate. This rule is consistent with the policy comparison, where reputation produces nearly the same long-run welfare as the combined portfolio at much lower cost.

11.2 Recommendation 2: Switch to Portfolio Governance under High Maintenance Pressure

When maintenance pressure is high, free-riding reduction alone is not enough. The critical-infrastructure simulations show that the combined portfolio creates the largest pressure reduction and the highest deterministic welfare, although its policy cost is higher. For infrastructure-like public goods, decision makers should combine incentive correction with backlog reduction and direct budget support when pressure relief is the dominant objective.

Operationally, a portfolio policy is justified when two signals appear together: the free-riding ratio remains high, and the pressure trajectory continues to rise after incentive correction. In that case, a pure reputation or penalty mechanism may improve effort without removing accumulated backlog. The model supports a staged design: correct incentives first, then add pressure-relief instruments if the stock–pressure system remains fragile.

11.3 Recommendation 3: Avoid One-Size-Fits-All Policy

Policy choice should depend on scenario conditions. In small-volunteer settings, heavy intervention may lower net welfare because policy cost can exceed additional benefit. In rapid-growth settings, subsidy and targeted incentive correction can help scale contribution, while matching alone should be monitored because it may leave free riding and pressure unresolved. In burnout-prone settings, simply forcing more contribution may increase burden; a balanced portfolio with support and pressure relief is more appropriate.

The Pareto results further suggest that portfolios should be compared by welfare, cost, free-riding reduction, fairness, pressure, and stability. In practice, the model can be used as a screening workflow: identify the dominant risk, apply low-cost incentive correction when the system remains stable, and escalate to a mixed portfolio only when pressure or reliability risk dominates.

Table 20: Scenario-based managerial decision rule suggested by the DSPF experiments.

Scenario type	Dominant risk	Suggested governance response
Small volunteer	Low pressure but uneven contribution	Use light reputation or matching; avoid costly heavy intervention.
Rapid growth	Demand expansion and pressure accumulation	Support scalable contribution through subsidy or matching, then monitor pressure.
Critical infrastructure	High public value and maintenance burden	Use reputation or penalty for incentive correction; add portfolio tools if pressure dominates.
Burnout-prone	High cost, high pressure, and contributor fatigue	Combine incentives with backlog reduction and budget support rather than only forcing effort.

12 Team Member Contributions

This study was completed independently. The author was responsible for problem interpretation, mechanism design, mathematical modeling, computational modeling, numerical experiments, figure generation, result interpretation, and final manuscript writing.

13 Discussion: Strengths, Limitations, and Extensions

13.1 Strengths

The model has five main strengths. First, it matches the problem requirements by including multiple agents, dynamic stock, natural decay, individual benefit, contribution cost, individual rationality, the social-planner benchmark, and intervention mechanisms. Second, it separates mechanism modeling from empirical claims through explicit synthetic labeling and parameter audit. Third, it compares Nash-style individual rationality and the planner benchmark in the same state space, making the free-riding gap measurable. Fourth, policy evaluation is multi-criteria rather than single-objective. Fifth, sensitivity, perturbed-scenario robustness, behavioral-response robustness, solver-damping checks, Monte Carlo, and Pareto analyses reduce dependence on one parameter setting or one calibration.

13.2 Limitations

The main limitation is the synthetic-only design. The model is suitable for mechanism exploration, but it is not an estimate of free riding in a specific real-world system. Agent utility is stylized and excludes social norms, communication, network effects, legal constraints, and repeated-game learning. The Nash-style solution is a converged numerical individual-rational response benchmark, not a proven unique Nash equilibrium. The robustness checks show that under-provision and the representative policy ranking are stable under the tested perturbations, while also identifying penalty-related weights that are inactive in the no-policy baseline. The social-planner benchmark is stage-wise and should not be read as an infinite-horizon global optimum. Policy-cost coefficients are synthetic accounting weights, so cost-effectiveness conclusions are comparative within the stated model rather than fiscal estimates. Practical implementation would require institutional and empirical validation.

13.3 Future Work

Future work can calibrate the model with real public-goods data when reliable observations are available, include network interactions among agents, model learning and reputation accumulation over repeated periods, and add explicit institutional constraints. A hybrid empirical-simulation strategy would be natural once trustworthy data on contribution, public-good quality, and maintenance cost are available. Possible calibration proxies include open-source maintenance records, community service participation logs, infrastructure maintenance budgets, or environmental governance indicators, but such proxies should be used cautiously because they observe only part of the public-good mechanism.

13.4 Reproducibility and Auditability Note

The numerical study is reproducible at the level of scenario definitions, random seed, policy settings, parameter-audit tables, and displayed precision. All scenarios are synthetic, the random seed is fixed at 20260614, and the same baseline parameter set is used for benchmark comparison, policy evaluation, sensitivity analysis, fine-grid response surface, critical-infrastructure robustness tests, Monte Carlo experiments, and Pareto search. Behavioral-response experiments perturb calibration weights jointly and one at a time; the damping experiment varies the numerical update weight around the baseline. In the scenario-perturbation table, the critical-infrastructure population is fixed before low-decay, high-decay, low-cost, high-cost, and demand-shock variants are applied. Runtime is recorded only as an audit item because it varies by machine and figure-regeneration setting. The substantive reproducibility claim rests on fixed seed, explicit scenario definitions, saved CSV outputs, and rounded table values; independent reruns are expected to differ only at machine

precision. This auditability supports internal consistency of the mechanism study, but it does not make the synthetic scenarios empirical measurements.

14 Conclusion

This paper develops a simulation-based dynamic multi-agent model for free riding in public-good provision. The model links heterogeneous agents, contribution, public-good stock, maintenance pressure, demand feedback, a Nash-style marginal-response fixed-point proxy, the stage-wise social-planner benchmark, and policy intervention. Under controlled synthetic settings, the numerical results show that decentralized individual rationality under-provides the public good relative to the stage-wise planner benchmark. The burnout-prone scenario has the largest free-riding gap, under-provision ratio, and welfare-loss ratio, while the critical-infrastructure scenario has the largest one-step maintenance-pressure gap.

Policy experiments indicate that well-matched interventions can reduce free riding and improve welfare, but their effects depend on scenario conditions and differ in cost, pressure relief, fairness, and stability. Reputation is cost-effective in the critical-infrastructure scenario; the combined portfolio is more suitable when maintenance pressure and system protection are central concerns. Sensitivity, perturbed-scenario robustness, behavioral-response robustness, solver-damping checks, Monte Carlo, and Pareto analyses support the main qualitative conclusion: public-good governance should not rely on a single instrument, and policy recommendations should be scenario-specific.

The practical value of the DSPF framework is diagnostic. It provides a reproducible workflow for asking whether a public-good system is mainly constrained by weak private incentives, accumulated maintenance pressure, excessive policy cost, or unequal contribution burden. All conclusions remain conditional on the synthetic simulation design. The model does not claim empirical calibration; it offers a mechanism-oriented framework for studying how free riding emerges dynamically, how calibrated response rules can be audited, and how policy instruments can be compared systematically.

References

- [1] Samuelson, P. A. (1954). The pure theory of public expenditure. *The Review of Economics and Statistics*, 36(4), 387–389.
- [2] Olson, M. (1965). *The Logic of Collective Action: Public Goods and the Theory of Groups*. Harvard University Press.
- [3] Bergstrom, T., Blume, L., and Varian, H. (1986). On the private provision of public goods. *Journal of Public Economics*, 29(1), 25–49.
- [4] Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859), 1243–1248.
- [5] Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press.
- [6] Fudenberg, D., and Tirole, J. (1991). *Game Theory*. MIT Press.
- [7] Mas-Colell, A., Whinston, M. D., and Green, J. R. (1995). *Microeconomic Theory*. Oxford University Press.
- [8] Andreoni, J. (1990). Impure altruism and donations to public goods: A theory of warm-glow giving. *The Economic Journal*, 100(401), 464–477.
- [9] Fehr, E., and Gächter, S. (2000). Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4), 980–994.
- [10] Nowak, M. A. (2006). Five rules for the evolution of cooperation. *Science*, 314(5805), 1560–1563.